BenGER: Benchmarking LLM Systems on Subsumption-Based Legal Reasoning in German Law

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Abstract

We introduce BenGER (Benchmark for German Law), a benchmark and dataset for subsumption-based legal reasoning in German law. It combines 596 exam-style free-text case tasks across several levels of legal education with 531 short doctrinal reasoning tasks. A controlled subset adds timed human solutions, written both unaided and in human-AI co-creation. We evaluate 12 contemporary LLM systems: closed flagship, efficiency-oriented, and open-weight. A rubric-aligned LLM-as-a-Judge scores them. We cross-validate it against a multi-rater human layer (three blind reviews per solution) and across six LLM judges. Closed-flagship systems lead across all three corpora. Human-AI co-creation measurably improves on unaided human work. The judge tracks human grading at Pearson r=0.76 and Cohen's κ=0.60. Rankings are stable across judges. Two judges from independent providers clear the Calderon single-reviewer replacement bar on human-authored solutions.

1 Introduction

Evaluating legal reasoning is hard. Even human experts grade the same answer very differently. Hufeld (Hufeld, 2024) shows that identical answers in German law exams receive markedly different scores depending on the grader. Doctrinal analysis is interpretive, so this is no surprise. But it is a problem for benchmarking large language models (LLMs). If human evaluation is itself noisy, what does it mean to benchmark legal reasoning?

Legal NLP benchmarks have evolved fast. They moved from classification (Chalkidis et al., 2022; Guha et al., 2023) to reasoning (Fei et al., 2024; Fan et al., 2025; Jia et al., 2025; Shi et al., 2026) and retrieval-augmented evaluation (Pipitone and Houir Alami, 2024; Butler and Butler, 2026). But most still emphasise short-form QA, common-law jurisdictions, or stable ground-truth labels. Open-ended subsumption-based German legal reasoning under noisy expert judgement remains underserved.

We address these gaps with a benchmark centered on subsumption-based legal reasoning, the core reasoning paradigm in German law (Section 2) and, more broadly, in many civil-law jurisdictions worldwide.

We present the BenGER (Benchmark for German Law) dataset[1] for evaluating LLMs on subsumption-based legal reasoning in German law. It has three components. Together they cover 596 exam-style free-text case tasks across different stages of legal education and 531 short doctrinal reasoning tasks. It also adds human baselines, both traditional legal work and human-AI co-creation, on a controlled Benchathon validation subset. Figure 1 summarises the three components and the generation-plus-evaluation pipeline. Generation, annotation, and evaluation ran end-to-end on the BenGER platform (Nagl and Grabmair, 2026), the open-source web application we built earlier for benchmarking legal-domain LLM systems.

How do we evaluate free-text reasoning when the human judgement is itself noisy? We adopt a hybrid framework. An LLM-as-a-Judge, aligned with German grading rubrics, does the scoring. We cross-validate it against three blind human reviews per solution, plus an author-informed creator review as a reference signal. We treat evaluation as a distribution over plausible expert assessments, not a single deterministic label.

We make three contributions. (i) A benchmark for subsumption-based legal reasoning in German law. (ii) An empirical evaluation of 12 contemporary LLM systems – closed flagship, efficiency-

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[1]Full code and dataset are available at https://github.com/SebastianNagl/benger-platform.

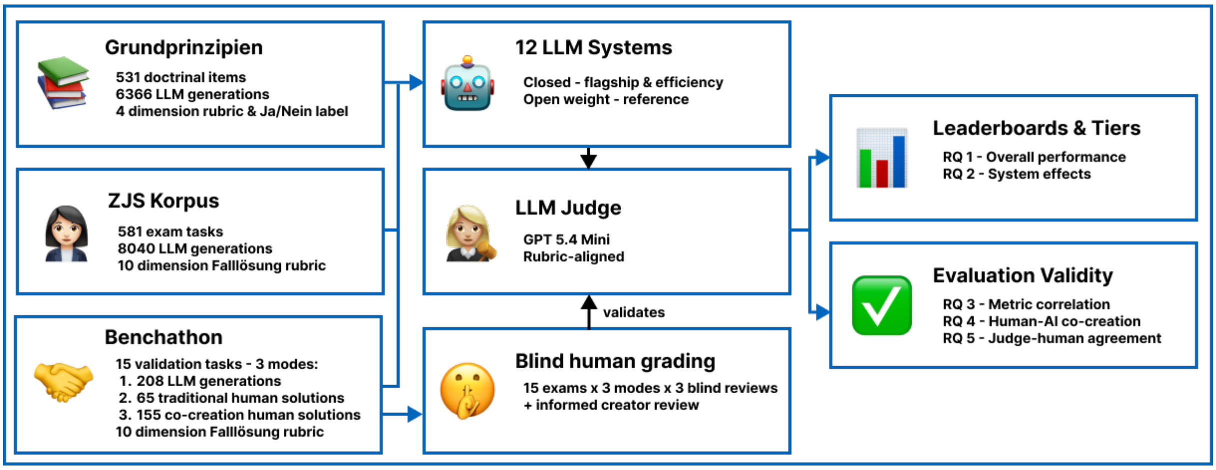

Figure 1: Overview of the benchmark and evaluation pipeline. Three German-law corpora – the published ZJS exam corpus, the short Doctrinal Principles question set, and the controlled Benchathon validation subset with human solutions in two working conditions – are answered by a panel of LLM systems and scored by a rubric-aligned LLM judge. The Benchathon subset is additionally graded by a blind human reviewer pool (with one author-informed creator review per solution); the resulting human distribution is used to validate the LLM judge.

oriented, and open-weight – alongside human and co-creative baselines. (iii) An evaluation methodology that accounts for the inherent variability of legal grading.

We organise the empirical analysis of the full BenGER dataset around five questions. **RQ1 (Performance):** how well do contemporary LLM systems solve open-ended subsumption-based German legal reasoning tasks, and how do they compare to human baselines? **RQ2 (System-tier effects):** how does performance vary across system tiers and open-/closed-weight access modes? **RQ3 (Evaluation validity):** how well do generic automatic metrics correlate with rubric-based legal evaluation? **RQ4 (Human-AI co-creation):** does AI-assisted drafting improve human performance, and how does it compare to standalone LLM systems? **RQ5 (Grading reliability):** how closely do LLM-judge evaluations align with the empirical distribution of human grades, and are their deviations comparable to inter-rater variability?

2 Subsumption-based legal reasoning

Subsumption is the core reasoning paradigm of German civil-law doctrine (Larenz and Canaris, 1995; Möllers, 2023). Common-law reasoning proceeds through analogy to precedent. German analysis works differently. It derives outcomes from codified statutory norms through a fixed four-step scaffold: *Obersatz* (the norm hypothesis to be examined), *Definition* (interpretation of the norm's elements), *Subsumtion* (anchoring the concrete facts under those elements)[2], and *Ergebnis* (the resulting legal consequence). The scaffold is the German articulation of the legal syllogism that underlies statutory reasoning across civil-law jurisdictions; what is Germany-specific is its rubric-graded written form, not the reasoning operation itself. This is the *Gutachtenstil*, the opinion style in which the conclusion follows the reasoning. It differs from the result-first *Urteilsstil* used in judicial decisions. German legal education, state examinations, and entry-level practice are all trained and assessed in this style. Grading targets the written *Falllösung* (case solution) end-to-end, not the final answer alone. The rubric rewards how each norm element is anchored in the specific facts.

This makes subsumption a structurally demanding generation task. A system must identify the governing norm, articulate its elements, anchor every element in the facts, and conclude. Surface fluency can mask a missing *Subsumtion* step. But the same structure makes the task unusually amenable to rubric-based evaluation. Each step in the scaffold maps to an identifiable part of the answer, not to a global quality impression. English-language legal NLP benchmarks built on classification, multiple-choice, or precedent-style reasoning (Chalkidis et al., 2022; Guha et al., 2023) do not exercise this scaffold. German terms are glossed

[2] *Subsumtion* carries a broad and a narrow sense. Broadly, it names the entire norm-application method – the sense in the paper title. Narrowly, it names the single fact-anchoring step of the scaffold – the sense used for the scaffold step and the rubric dimension of the same name.

at first use throughout; a consolidated glossary is in Appendix Q.

3 Related work

Legal NLP benchmarking has moved from classification to reasoning. Early baselines were LexGLUE (Chalkidis et al., 2022) (seven English legal datasets) and LegalBench (Guha et al., 2023) (162 mostly short-form US-federal-law tasks). Recent work targets jurisdiction-specific reasoning: LawBench (Fei et al., 2024) and LexEval (Li et al., 2024) for Chinese civil law, KBL (Kim et al., 2024) for Korean bar-exam questions, LAR-ECHR (Chlapanis et al., 2024) for European Court of Human Rights argument chains, KOBLEX (Lee et al., 2025) for provision-grounded multi-hop reasoning in Korean law, GreekBarBench (Chlapanis et al., 2025) for free-text Greek bar exams, NitiBench (Akarajaradwong et al., 2025) for Thai financial and tax law, PLawBench (Shi et al., 2026) for rubric-graded Chinese legal practice across 13 practical scenarios, and LaborBench (Hariri and Ho, 2025) for US state labor-law statutory simplification. A complementary line targets retrieval-augmented evaluation, LegalBench-RAG (Pipitone and Houir Alami, 2024) and Legal RAG Bench (Butler and Butler, 2026), and discrepancy-based auditing of LLM legal reasoning, CLAUSE (Choudhury et al., 2026). The closest precedent is LEXam (Fan et al., 2025): 7,537 questions across 340 law exams in English and German, with an ensemble LLM-as-a-Judge validated against human experts. But its rubric is not the *Obersatz-Definition-Subsumtion-Ergebnis* scaffold central to German *Falllösung* grading (Section 2). And it does not couple its free-text items to the dimension-level grading criteria used in German law school and state-exam evaluation.

Free-text legal evaluation increasingly relies on LLM-as-a-Judge approaches. These raise concerns over bias, calibration, and alignment with expert judgement. German legal education shows how variable human grading already is: when Hufeld (Hufeld, 2024) had 23 graders score 15 beginner-level law exams, the average range between the lowest and highest grade per answer was 6.47 points on the 0-18 scale, and only 42% of grades fell within ±1 point of the answer-specific mean. Legal grading is therefore better treated as a distribution over plausible expert judgements than as a single ground truth, which motivates hybrid methods that combine human expertise with calibrated LLM-based scoring.

Resources for German legal NLP remain limited. GerLayQA (Büttner and Habernal, 2024) provides ~21,000 lay questions paired with lawyer answers, but focuses on QA rather than subsumption. GerLeRB (Weber et al., 2025) targets legislative retrieval. Our prior BenGER platform (Nagl and Grabmair, 2026) provides an end-to-end framework for German legal case analysis and subsumption-based reasoning workflows, on which the present dataset was built. German reasoning grounded in doctrinal subsumption under statutory law remains under-served, especially for open-ended, structured analysis under noisy human judgement. The BenGER dataset addresses this gap. It combines a 596-task exam-style corpus and 531 short doctrinal reasoning items with a controlled Benchathon validation subset, a German doctrinal rubric, and a multi-rater cross-validation protocol.

4 Dataset

The BenGER dataset draws on three components. They cover the main forms of legal reasoning in German legal education. Two are exam-style case corpora: a fact pattern is given, and a structured *Falllösung* is expected. The third is a set of short doctrinal reasoning items, probing core principles without the full case-analysis scaffolding.

Only the first component builds on pre-existing material. The ZJS corpus contributes 581 publicly available exam-style cases from the *Zeitschrift für das Juristische Studium* (Journal for the Study of Law)[3]. They cover civil, criminal, and public law. They span early undergraduate exercises through first and selected second state-examination materials. Each is paired with an expert-written reference solution. The cases and reference solutions pre-exist this work as journal publications; our contribution is the benchmark artifact: selection, machine-readable task-reference pairs, per-item IP clearance (§Ethical considerations). The Benchathon corpus adds 15 newly collected exam-style tasks at an intermediate difficulty level, more demanding than introductory exercises but less complex than full first-state-exam problems. The tasks and their reference solutions were authored for this benchmark by the domain experts on the author team (the task authors of Appendix F). It com-

[3] https://www.zjs-online.com/

prises 220 human solutions in total (65 traditional, 155 human-AI co-creation; about 15 per task).[4] The Doctrinal Principles corpus (*Grundprinzipien*) contributes 531 short doctrinal reasoning items – question-answer pairs with explanatory reasoning – on core principles, with recent case law where appropriate. The items were newly written for this benchmark by a subset of the authors with the relevant domain expertise and, like Benchathon, were unpublished prior to this work. The three play distinct roles. ZJS supports at-scale automated evaluation. Benchathon supports the human baseline and judge-validation layer. Doctrinal Principles probes the same reasoning at shorter form.

The benchmark is for evaluation only. We do not define a train/test split. The objective is comparative evaluation of deployed LLM systems and human baselines. Pretraining-contamination considerations across the three corpora are discussed in §Limitations.

4.1 Benchathon task design and human solution collection

Benchathon participants solved tasks in a controlled annotation environment. Each task was randomly assigned to one of two conditions. In the *traditional* condition, participants could use statutory texts, legal databases, literature, and internet research. In the *co-creation* condition, they could additionally use any LLM(s) of their choosing – commercial or open-weight, in any combination – through their own accounts. We did not log which tool a participant used. The co-creation arm therefore measures an upper bound on what human-LLM teams currently achieve with off-the-shelf assistance, not a controlled comparison against any specific system. Only the final written solution is evaluated, mirroring legal examination practice. Each task had a two-hour time limit. Participant expertise was measured objectively (reported legal grades and qualifications) and subjectively (self-assessed competence on a Likert scale).

[4]The corpus is named after the in-person Benchathon event at which participants produced these solutions and the validation human grading. Participants were predominantly law students at various stages of legal training, plus a smaller group of *Referendare* (post-state-exam legal trainees), recent graduates, and a small layperson cohort; full composition in Appendix E.

5 Experimental setup

5.1 Generation

We evaluate 12 contemporary LLM systems: 6 closed-weight through provider APIs (OpenAI, Anthropic, Google) and 6 open-weight through a managed inference provider (DeepInfra). Per-system providers, snapshots, observed temperatures, token-budget settings, truncation counts, and the short aliases used in the body are in Table 10 (Appendix B). Each task received at least one generation per system. Flagship and large-context systems received additional repetitions where capacity allowed. All systems generated the Benchathon subset. On ZJS, all 12 systems covered up to 581 published exams.[5]

All systems received the same German-language system and instruction prompts (Appendix J + K). Each returned a single JSON object containing the full case solution (*Falllösung*). The instruction prompt mirrors the *Gutachtenstil* scaffold (Section 2). We used provider-recommended temperatures and token budgets (temperature 0.0-1.0, max output 8,000-16,000 tokens). Generations that were malformed or shorter than 200 output tokens after a bounded retry budget were excluded. Truncation in kept generations falls mostly on the smaller open-weight systems, with a few on Gemini-3.1-Pro (Table 10).

5.2 Black-box system evaluation

We evaluate LLM systems through provider APIs in a black-box setting. Our results therefore measure end-to-end system behaviour, not isolated base-model capability. We do not enable retrieval or tool-use endpoints. Components we cannot disable operate by default – notably content filters and, on some products, internal routing between submodels. This reflects realistic deployment conditions. But it limits causal claims about underlying architectures; for open weights we test the reasoning mechanism directly with a same-base-model Thinking-vs.-Instruct toggle (Appendix P).

5.3 Automatic metrics

Per generation we compute BLEU (Papineni et al., 2002), ROUGE (Lin, 2004), METEOR (Banerjee and Lavie, 2005), chrF (Popović, 2015), BERTScore (Zhang et al., 2020), MoverScore

[5]Individual questions were occasionally flagged as prohibited content by the provider's safety filter, leaving a small handful of per-system coverage gaps below the corpus total.

(Zhao et al., 2019), and a Sentence-BERT cosine similarity (Reimers and Gurevych, 2019) against the expert reference solution. Per-system means appear in the main leaderboard tables. Per-generation values are correlated against the LLM-judge rubric score in RQ3.

5.4 LLM-as-a-Judge evaluation

For rubric-based evaluation on ZJS and Benchathon, we use a specialised LLM-as-a-Judge prompt aligned with German legal grading practice. The judge receives the task, the reference solution, and the solution to be evaluated. It scores ten dimensions: result correctness (*Ergebnisrichtigkeit*), issue identification (*Vollständigkeit*), legal grounding (*Rechtsgrundlagen*), doctrinal knowledge (*Rechtskenntnis*), subsumption quality (*Subsumtion*), problem depth (*Schwerpunktsetzung*), methodological structure (*Methodischer Stil*), organization (*Gliederung*), terminology (*Sprache*), and formal correctness (*Formalia*). Dimension scores aggregate to a 100-point raw score, mapped to the German 0-18 grade-point scale. A generation *passes* when its grade-point score reaches *ausreichend* – ≥ 4 on the 0-18 scale, a raw score of ≥ 50/100 – following the German *Staatsexamen* (state law examination) convention.[6] This is the boolean in the *Pass* columns of Table 1 and in the pass-rate aggregates throughout. The rubric operationalises the *Gutachtenstil* scaffold (Section 2): each step in the *Obersatz-Definition-Subsumtion-Ergebnis* schema is reflected in a dedicated dimension (mapping and /100 weights in Table 8, Appendix A); scaffold-step dimensions carry 90 of the 100 raw points, and a leaderboard on those seven dimensions alone reproduces the full-rubric system ranking exactly on both *Falllösung* corpora (Spearman ρ = 1.00/1.00; Table 9). The per-dimension scores in Figure 5 therefore track identifiable components of the reasoning process. Grundprinzipien, the short-form Doctrinal Principles corpus, uses a smaller 4-dimension rubric (result correctness, legal knowledge, subsumption, clarity). It is not mapped to the grade-point scale, since it does not follow the full *Gutachtenstil* scaffold.

The judge is blind to solution origin and receives inputs in a fixed (task, reference, candidate) order. The primary judge is GPT-5.4-mini[7]. Judge-based evaluation may itself be model-dependent. We therefore cross-validate the judge against a multi-rater human-grading layer on the Benchathon subset. We also run two stability checks on that subset: *within-judge stochasticity* (k=3 re-runs of GPT-5.4-mini) and *between-judge bias* (additional evaluations by Opus-4.7, Gemini-3.1-Pro, DeepSeek-V4-Pro, Qwen3.5-397B-A17B, and Sonnet-4.6). Both are reported in RQ5. Outputs conform to a predefined JSON schema. Non-conforming outputs are retried, and reported as evaluation failures if they cannot be parsed.

5.5 Human evaluation

Human evaluation runs on a controlled subset of 45 Benchathon solutions: one traditional human, one human-AI co-created, and one LLM-generated solution per Benchathon task. Human submissions are the median-expertise participant per legal area (*Bereich*). LLM submissions are drawn through a tier-balanced rotation across closed-flagship, efficiency-oriented, and open-weight reference systems. Each selected solution receives three independent blind reviews and one author-informed creator review (180 reviews total). Reviewers grade against the same rubric as the LLM judge and form the IRR pool used to validate it. Creator reviews are excluded from IRR, since task authors have privileged knowledge. The selection procedure, the seven-grader roster, and the $m = 5$ blind subset (the graders who performed blind reviews; each solution is reviewed by three of them) are described in Appendix F.

We report descriptive agreement statistics – Spearman, Pearson, MAE, Cohen's κ (Cohen, 1960), ICC(2,1) and ICC(2,k) (Shrout and Fleiss, 1979; Koo and Li, 2016) – alongside the Calderon alt-test (Calderon et al., 2025) in RQ5. Each answers a distinct question – rank vs. absolute agreement, error magnitude, the pass/fail threshold decision, and the reliability ceiling of the human pool itself (per-statistic rationale in Appendix I). The Calderon alt-test is the primary criterion: it is purpose-built to decide whether the judge can stand in for a blind reviewer. For

[6] The ≥ 4 pass mark is statutory: the federal grading ordinance (Bundesministerium der Justiz, 1981) fixes *ausreichend* (4-6 of 18) as the lowest passing grade on a deliberately severe scale (≈30% fail the first state examination). The raw-to-grade conversion is non-linear – grade 4 requires ≥ 50/100 raw points (table in Appendix L).

[7] Configuration in Appendix M, prompt in Appendix L. The ARR-reviewed version used GPT-5-mini as primary judge; the final version re-scored every generation with GPT-5.4-mini and extended cross-validation to six judges – tier-level conclusions unchanged, absolute scores a calibrated ordering (Limitations).

the alt-test we follow the blind-pool procedure at $\varepsilon = 0.15$ (Calderon's skilled-annotator tier), with Benjamini-Yekutieli FDR at $q = 0.05$ across the m blind reviewers. We also report the single-expert variant against the un-blind creator grade at $\varepsilon = 0.20$ (with $\varepsilon = 0.15$ as a sensitivity check). Following Calderon's single-expert case, this variant treats the un-blind creator grade as the sole reference: each blind reviewer and the judge are scored by absolute distance to that single creator grade on every shared solution, and ω is the fraction of the m blind reviewers the judge matches or beats. So it needs one creator grade per solution, not a pool of expert votes. The judge passes when the winning rate $\omega \geq 0.5$. The primary judge is GPT-5.4-mini; we additionally re-score the same picks with Opus-4.7, Gemini-3.1-Pro, DeepSeek-V4-Pro, Qwen3.5-397B-A17B, and Sonnet-4.6 (per-judge alt-test in Tables 23 and 24). The central question is not whether the LLM judge reproduces a single human grade. It is whether its deviation is comparable to that among human graders.

6 Results

Unless otherwise noted, scores in this section are the LLM judge's rubric scores, as 0-100 raw points or 0-18 German grade points. The human-grading layer covers the 45-pick validation design: one author-informed creator review and three blind reviews per pick (180 reviews total). The released analysis pipeline regenerates every number reported below from the data files published with the paper.[8]

6.1 RQ1: Overall performance of LLM systems and humans

Table 1 gives the unified per-system view across the three corpora; per-corpus full leaderboards with surface-metric columns are in Appendix A. On Benchathon, the top of the table is a closed-flagship cluster – Opus-4.7, Gemini-3.1-Pro, and GPT-5.4 in the upper 60s on the 0-100 judge scale, with pass rates above 90% – followed by an efficiency band in the upper 50s to lower 60s. The strongest open-weight system, DeepSeek-V4-Pro, trails the top of the flagship cluster by roughly 13 raw points. The open-weight tier spans a wide range; the nominal flagship Qwen3-235B sits 28 raw points behind the lead and underperforms its size here, despite no truncation. The same ordering broadly reproduces on ZJS, where GPT-5.4 leads and the weakest open-weight system, Llama-4, trails the field; per-system scores stay in Table 1.

Within-task variance is substantial. Under the GPT-5.4-mini judge, per-system standard deviations across the Benchathon tasks span roughly 8-17 raw points across all tiers. So single-task performance is a noisy signal, even for top-tier systems. On Benchathon, the human cohort scores in the mid-range of the LLM distribution under both conditions, shifted upward under co-creation (see human rows in Table 1 and Figure 2 in RQ4).

The same flagship cluster leads on the short-form Doctrinal Principles corpus, with Sonnet-4.6 joining it. Doctrinal Principles items carry a Ja/Nein (yes/no) decision label; across systems, decision accuracy ranges from 69% to 83%.

6.2 RQ2: Scaling and system effects

Closed flagship systems from OpenAI, Anthropic, and Google occupy the top of every per-corpus leaderboard. Efficiency-tier closed systems and the strongest open-weight system (DeepSeek-V4-Pro) form an overlapping mid-band. Closed-API dominance narrows below the flagship tier. By tier, the flagship-vs-open-weight gap is 21 raw points on Benchathon and narrows to 16 on ZJS. The closed-vs-open-weights gap shrinks similarly, from 20 to 13 raw points. The same direction holds on Doctrinal Principles. The flagship-vs-open-weight ordering therefore survives all three task formats and the 4-dimension rubric variant. We frame these as system-level effects, not model-level ones. Black-box API access entangles the open-closed split with provider-side prompts, decoding, post-processing, and possible retrieval or tool use (see Limitations).

6.3 RQ3: Evaluation validity - automatic metric correlation

Across all three corpora, lexical surface metrics correlate with the judge's rubric raw score more strongly than embedding-based metrics do. This runs partly against the common expectation that semantic-embedding metrics dominate lexical ones on long-form legal text. On Benchathon the lexical metrics lead: METEOR (r=+0.61), ROUGE (r=+0.68), and BLEU (r=+0.58). MoverScore (r=+0.59) is the one embedding metric on par with them. BERTScore (r=+0.47) and sentence-embedding similarity (r=+0.33) trail. The same ordering broadly holds on ZJS and

[8] https://doi.org/10.5281/zenodo.20489788

Table 1: Main leaderboard across the three corpora. Per-system mean LLM-judge raw score (0–100, mean ± half-width of a 95% CI: percentile bootstrap $B = 2000$ on Benchathon for small per-system n, analytic $1.96 \cdot \mathrm{SE}$ on ZJS and Doctrinal Principles) and pass rate. Doctrinal Principles also reports Ja/Nein (yes/no) decision accuracy. Systems are ranked by the mean of their available per-corpus raw scores. The two human-baseline rows use the LLM judge's grades of human-written Benchathon solutions.

	Benchathon			ZJS			Doctrinal Principles			
System / Group	n	Raw	Pass	n	Raw	Pass	n	Raw	Pass	Acc.
Opus-4.7	15	69.3 ± 5.6	93%	652	58.0 ± 1.0	74%	531	83.1 ± 2.0	86%	81%
Gemini-3.1-Pro	15	68.3 ± 8.4	93%	740	58.8 ± 1.0	77%	531	83.2 ± 1.8	87%	83%
GPT-5.4	30	68.2 ± 3.5	97%	586	60.4 ± 1.1	78%	531	80.3 ± 2.1	82%	79%
Sonnet-4.6	15	58.9 ± 6.0	73%	782	49.9 ± 0.8	46%	531	81.6 ± 2.1	83%	81%
Gemini-3.1-Flash-Lite	15	63.5 ± 5.8	87%	579	44.6 ± 0.8	26%	531	73.3 ± 2.3	76%	76%
GPT-5.4-mini	15	58.6 ± 5.5	73%	581	49.7 ± 0.9	46%	531	71.8 ± 2.4	72%	71%
DeepSeek-V4-Pro	15	56.1 ± 5.9	67%	609	49.8 ± 1.0	44%	531	73.5 ± 2.3	73%	73%
DeepSeek-V4-Flash	15	49.3 ± 3.8	47%	582	45.9 ± 0.8	30%	530	70.7 ± 2.4	73%	69%
Qwen3.5-122B	15	49.4 ± 4.9	40%	863	41.3 ± 0.7	19%	526	69.8 ± 2.4	75%	78%
Qwen3.6-35B	15	42.4 ± 7.2	27%	829	37.9 ± 0.7	11%	531	69.5 ± 2.5	69%	71%
Llama-4	15	37.8 ± 4.1	20%	581	36.1 ± 0.6	4%	531	70.1 ± 2.3	75%	77%
Qwen3-235B	28	41.3 ± 4.3	18%	656	35.3 ± 0.7	5%	531	67.3 ± 2.5	69%	73%
Human (all)	220	61.1 ± 2.1	77%	—	—	—	—	—	—	—
traditional	65	50.0 ± 3.5	52%	—	—	—	—	—	—	—
co-creation	155	65.7 ± 2.2	88%	—	—	—	—	—	—	—

Doctrinal Principles (Appendix H: Table 15, Figure 4). This fits the highly templated structure of doctrinal German exam writing, where reference and high-quality system answers share substantial n-gram material around statutory citations and the *Obersatz-Definition-Subsumtion-Ergebnis* scaffolding.

At the per-system level, the ordering shifts. Embedding-based metrics rank systems at least as well as lexical ones. On Benchathon, ROUGE leads (Spearman +0.89), but BERTScore (+0.80) and MoverScore (+0.81) outrank BLEU (+0.64) and METEOR (+0.60). On ZJS, BERTScore is the single strongest ranker (+0.95). A leaderboard built on a strong surface or embedding metric would broadly preserve the judge's coarse-tier ordering, but disagree on the mid-tier. This motivates a calibrated LLM judge as the primary scoring instrument.

A natural concern is that the lexical advantage could reflect a judge-specific preference for reference-like text rather than a property of the task. Two analyses on the 45-pick validation subset test this against the mean blind-human grade and say otherwise: the lexical correlations persist under human anchoring (r≈+0.72 pooled), and the residual diagnostic – correlating (judge – mean blind grade) with each lexical metric – comes out *negative* for all three: relative to blind human graders, the judge *under*-rewards high-overlap solutions, the opposite sign of a reference-style bias (Table 16 in Appendix H, with per-metric values, length-controlled partials, and small-n caveats).

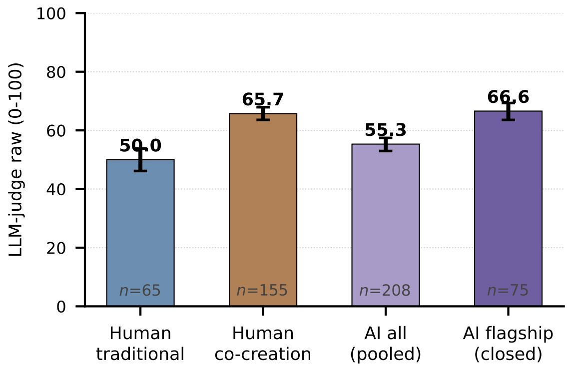

Figure 2: Mean LLM-judge raw score (0-100) per group on the Benchathon validation subset, with 95% bootstrap CIs on each group's mean. The two AI bars pool per-generation scores across all evaluated LLM systems (*pooled*) and across the closed-flagship tier only (*flagship*).

6.4 RQ4: Human-AI co-creation under LLM-judge scoring

Co-creation solutions score higher than traditional ones on Benchathon – by +15.7 raw points on average, about +2.8 grade points on the 0-18 scale (Figure 2; participant-clustered 95% CI [+9.2, +22.2]). The gap holds within the 23 participants who completed both arms[9]. It repeats across all three legal domains. It is in fact a lower bound on the per-participant gain: less-experienced participants skipped traditional items more often than co-creation ones, leaving the traditional pool expertise-biased upward (full robustness arc in Appendix G). With that lift, the AI-assisted humans sit alongside the best standalone LLMs we evaluated. A TOST equivalence test rules out any difference of more than ±5 raw points against the closed-flagship tier (Table 1, Appendix G). Re-graded by blind human reviewers on the RQ5 validation subset, the gap shrinks to about 14 raw points. That is +2 points smaller than the LLM-judge estimate, a small absolute discrepancy. It is consistent with the GPT-5.4-mini judge's near-zero content-dependent calibration offset, and within the blind-pool's inter-rater spread (Limitations).

6.5 RQ5: Grading reliability

The reliability question is simple: can the LLM judge stand in for a blind human reviewer? We answer it in three stages: how closely the judge agrees with humans per solution, whether it clears the formal substitution bar, and what its disagreements look like when they happen.

Across the validation subset, the primary GPT-5.4-mini judge agrees with the per-solution mean of the blind reviewers at Pearson r=0.76, pass/fail Cohen's κ=0.60, and MAE 10.8 raw / 2.7 grade points. The per-solution offset is just -0.01 raw points. The reviewers themselves disagree by an average within-annotation max-min spread of 23.9 raw points, comparable to the several-grade-point range Hufeld (Hufeld, 2024) documents for German legal grading. The pool reaches ICC(2,k)=0.84. So the judge sits inside the cloud of human disagreement, not outside it (Table 17).

Per-solution agreement does not by itself show that the judge can *replace* a randomly drawn reviewer. The Calderon alt-test sets a stricter bar: $\varepsilon = 0.15$ for the blind grader pool (*skilled*) and $\varepsilon = 0.2$ for the single un-blind creator reference (*expert*), with Benjamini-Yekutieli FDR (§5.5). We apply it across six judges: the primary GPT-5.4-mini, plus Opus-4.7, Gemini-3.1-Pro, DeepSeek-V4-Pro, Qwen3.5-397B-A17B, and Sonnet-4.6.

For the blind pool, two judges clear the $\omega \geq 0.5$ replacement bar on the human-authored picks: GPT-5.4-mini at $\omega = 0.60$ and Opus-4.7 at $\omega = 0.60$. Sonnet-4.6 ($\omega = 0.40$) and Gemini-3.1-Pro ($\omega = 0.40$) fall one annotator-rejection short. DeepSeek-V4-Pro and Qwen3.5-397B-A17B fail at $\omega \leq 0.20$. At $m = 5$ these are point estimates; we read the two-judge pass as directional, not established (Limitations).

Against the single-expert creator reference, GPT-5.4-mini passes on human picks ($\omega = 0.60$) but fails on LLM picks ($\omega = 0.00$) (Table 17; per-judge ω, per-annotator p-values, and a stacked-bootstrap robustness check in Appendix N).

Agreement is highest on dimensions with structured scoring criteria and lowest on presentation and form (Figure 5). Several outcome and reasoning dimensions drop on LLM-generated solutions. Result correctness goes from r=0.76 (human-authored) to r=0.62 (LLM), with comparable or larger drops on Issue identification, Doctrinal knowledge, and Problem depth.

A single LLM judge re-run is more self-consistent than the human pool is internally. Re-running GPT-5.4-mini $k = 3$ times on the Benchathon LLM generations yields a within-cell standard deviation of 3.45 raw points. The blind reviewers' within-solution grades span a 23.9-raw-point max-min range. The two are different dispersion measures. But the judge's re-run noise is clearly far smaller than human grading disagreement.

6.6 Error analysis

On the two free-text *Falllösung* corpora, two dominant failure modes split by system tier. (i) *Long-but-shallow generations*: over 2,000 words, but scoring at most half marks on *Subsumtion* and problem depth. These concentrate in the flagship tier. The pattern is over-writing, which the rubric does not reward. (ii) *Wrong-norm framing generations*: methodologically well-formed but low on legal grounding and doctrinal knowledge. These concentrate in open-weight systems. The pattern is doctrinal mis-grounding, not mis-execution within the correct frame. The two modes are largely distinct (134 of the ~1,800 flagged ZJS generations fall in both). Per-mode counts and per-tier rates for all flagged modes on all three corpora are in Appendix O.

On the short-form Doctrinal Principles corpus, Ja/Nein accuracy follows the same tier ordering, and the wrong decision itself is the dominant error. Fact-bound *Subsumtion* and outcome correctness

[9] Participants who submitted both traditional and co-creation solutions.

are tightly coupled on every corpus, so reasoning-outcome divergence is rare but does occur. Where it occurs, the failure is in consistency rather than reasoning. The model reaches a wrong decision despite sound *Subsumtion*, or the right decision on flawed *Subsumtion*. Its stated answer is decoupled from its own reasoning – an inconsistency that Ja/Nein accuracy cannot capture but the rubric does.

7 Discussion

Across the three corpora, the top of the leaderboard is consistently Opus-4.7, Gemini-3.1-Pro, and GPT-5.4, with Sonnet-4.6 joining on Doctrinal Principles. The flagship-vs-open-weight gap is ~21 raw points on Benchathon (RQ2). It narrows on the longer ZJS corpus, and further on Doctrinal Principles. The strongest open-weight system, DeepSeek-V4-Pro, overlaps the efficiency-tier closed band on the longer corpora and closes on the flagships on the short one. So the practical question for German legal NLP is not whether open-weight systems can be used. It is whether the residual flagship advantage justifies the open- versus closed-weight access constraints (Appendix B).

Surface metrics (BLEU, ROUGE, METEOR) correlate with the rubric judge more strongly than embedding metrics do (RQ3), consistent with the templated *Obersatz-Definition-Subsumtion-Ergebnis* structure of German exam writing. A lexical-only leaderboard would keep the coarse tiers but disagree on the mid-tier. So the rubric judge remains the primary instrument, and surface metrics read as cheap correlates.

Human-AI co-creation lifts unaided humans into the closed-flagship band, with a magnitude caveat. AI-assisted human work scores +15.7 raw points ($\approx$ +2.8 German grade points) above unaided human work under the judge (RQ4). The gap holds within-participant and across the three legal areas. A TOST equivalence test renders it statistically indistinguishable from the closed-flagship LLM tier (Appendix G). Blind-human re-grading trims the headline modestly, but leaves its direction and the qualitative conclusion unchanged, with the residual within blind-pool inter-rater spread.

Two judges from independent providers clear the substitution bar at the point estimate (directional; see Limitations). GPT-5.4-mini tracks the per-solution blind-reviewer mean at Pearson r=0.76 and sits inside the human disagreement cloud (RQ5). On the Calderon alt-test at $\varepsilon = 0.15$, GPT-5.4-mini and Opus-4.7 both clear the $\omega \geq 0.5$ bar on human-authored picks ($\omega = 0.60$ and 0.60). The other four judges fall short. Rank agreement across the six judges is high: re-ranking the 12 systems under each cross-validation judge yields system-level Spearman ρ of 0.94-0.97 against the primary judge (Table 20). So the RQ1-RQ2 ordering survives any single judge's score bias. Two judges from different providers converge on the bar and on the reviewers they replace, which suggests same-provider bias is not a large confound. An Opus-anchored cross-check of the primary scores is a natural next step.

LLM judges are thus scalable raters alongside humans, not standalone ground truth. Judge-model choice is a deliberate calibration knob. Where the judge still diverges, the disagreement concentrates on presentation and on outcomes reached via unfamiliar reasoning paths. Agreement is highest on the outcome and issue-identification dimensions and lowest on formal correctness and terminology. It slips further on LLM-generated solutions.

8 Conclusion

We introduce the BenGER dataset for subsumption-based legal reasoning in German law. It combines two exam-style corpora (ZJS, Benchathon) with a short doctrinal-reasoning corpus (Doctrinal Principles), a Benchathon validation subset with human and human-AI baselines, a rubric-aligned LLM-as-a-Judge, and a multi-rater cross-validation protocol. That is 12 systems and 14614 generations on Benchathon, ZJS, and Doctrinal Principles, all run end-to-end on our previously released BenGER platform (Nagl and Grabmair, 2026). BenGER is released as an open evaluation framework with corpora, judge prompts, rubric, and analysis pipeline at https://github.com/SebastianNagl/benger-platform. The methodology travels: treating grading as a distribution over expert assessments and validating a judge by reviewer substitution apply to any expert-graded free-text domain, and the released pipeline is rubric-parameterized.

Limitations

Several limitations bound our findings:

All LLM systems are evaluated through hosted provider APIs in a black-box setting. Reported scores therefore reflect end-to-end deployed behaviour, not isolated base-model capability. We do not vary provider-side prompts, decoding configurations, retrieval, or tool use. We therefore make no claim about the contribution of individual architectural choices.

The dimension rubric and the LLM-judge prompt were co-developed for this benchmark, not adopted from independent prior work. We validate them as a pair against the multi-rater human-grading layer (RQ5). Its within-annotation inter-rater spread (5.2 grade points) sits within the range Hufeld (Hufeld, 2024) documents for German legal grading. Neither component has been validated in isolation.

The Benchathon subset comprises 15 tasks and a modest participant pool. This suffices for the per-task, per-system comparisons and the judge validation reported here. But it limits statistical power for fine-grained interaction effects, such as expertise-stratified co-creation analyses.

Judge bias is a known confounder in LLM evaluation (Zheng et al., 2023; Panickssery et al., 2024). We address it via the multi-rater agreement protocol on the Benchathon validation subset (RQ5) and the inter-judge agreement analysis. That analysis shows a clear strictness gradient, but pairwise rank agreement comparable to inter-human agreement.

Same-provider preference might occur, but its plausible size is limited. Four observations. (i) Cross-judge Spearman across all judges holds the system ordering across providers (system-level $\rho \geq 0.94$, Table 20). (ii) Judge-human Pearson is *lower* on the LLM half (r=0.64 vs. r=0.76), the opposite of same-family inflation. (iii) Calderon ω drops from 0.60 (human) to 0.00 (LLM), the same direction. (iv) The LLM-pick ω drop appears across all six judges tested – closed and open-weight, including the ones whose families lead the system rankings. So the lift is judge-model-broad, not family-aligned (Tables 21, 23, 24).

Per-annotator $n_j \in [7, 21]$ is well below Calderon's recommended 50-100. Every test therefore fell back to Wilcoxon, with reduced power. At this sample size the two-judge alt-test pass is a point estimate, not a robust result. The winning rate moves in steps of 0.20 at $m = 5$ reviewers. The 95% CI on the primary judge's ω spans [0.15, 0.95]. No stacked-bootstrap lower bound clears $\omega = 0.5$ for any judge (Appendix N). We therefore read the pass as directional, not established.

Absolute RQ1-RQ2 leaderboard scores carry the GPT-5.4-mini calibration offset (Table 21; measured against both the blind-reviewer pool mean and an independent creator-anchored reference in Appendix N.1). They should be read as a calibrated ordering, not human-pool estimates. Calderon's FAQ Q12 recommends per-domain ω, but per-area n_j would drop to 3-7, so we pool across the three legal areas (*Bereiche*: civil, public, and criminal law).

The rubric is grounded in German doctrinal grading conventions. It is not designed to generalise to common-law or non-German civil-law systems. Metrics and aggregate scores in the BenGER dataset should not be transferred to other jurisdictions.

ZJS material is publicly indexed. Provider models may have encountered some of it during pre-training. We have not run an explicit memorisation probe. The Benchathon and Doctrinal Principles corpora were unpublished at the time of all generation runs. They are not a clean contamination control: their orderings agreeing with the ZJS leaderboard is consistent both with low contamination and with contamination that scales proportionally across systems. What the agreement does bound is *catastrophic* differential contamination – a system whose ZJS rank were driven by memorisation, and that would underperform on the then-unseen Benchathon and Doctrinal Principles items, would visibly fall out of order on those corpora.

Per-domain CIs (RQ4), pairwise inter-judge correlations (RQ5), and error-analysis patterns are reported descriptively. We draw no inferential claims from individual cell-level comparisons.

The Benchathon and ZJS exam-style corpora are graded with the 10-dimension case-solution (*Falllösung*) rubric. The short-form Doctrinal Principles corpus uses a 4-dimension rubric (result correctness, legal knowledge, subsumption, clarity). Per-dimension comparisons across corpora are therefore not meaningful. Only aggregate raw scores, pass rates, and Ja/Nein (yes/no) accuracy (Doctrinal Principles only) should be compared across the three.

The Google efficiency-tier slot was generated with two distinct Google models: Gemini-3-Flash on the Benchathon subset, and Gemini-3.1-Flash-Lite (a separate efficiency product line, not a successor release of Flash) on the ZJS and Doctrinal Principles corpora. The model set was finalised between the two generation runs. We treat them as the same conceptual tier slot for tier-level comparisons and disclose the raw model ids in Table 10. Row-level cross-corpus comparison for this slot should be read with the model drift in mind. And tier-level claims should not be read as a per-model claim about either Flash or Flash-Lite alone.

Ethical considerations

Benchathon participants were recruited from a pool of law students and recent graduates who consented to the use of their written solutions and self-reported expertise data for research purposes. Solutions and self-reports are anonymized in the released BenGER dataset. Human grading is conducted by domain experts under the same consent regime. We release dimension-level rubric scores and aggregate statistics, not free-text feedback that could re-identify a participant. The Benchathon study design for the human and co-creation baseline was approved by an ethics committee. The LLM-as-a-Judge methodology in BenGER is intended for research-scale evaluation of legal-reasoning systems. It is not validated for use as a graded assessment instrument in legal education, or for deployment in legal practice.

The ZJS exams and reference solutions are copyright-protected. We have cleared IP rights for the vast majority of items, bundled directly with the released dataset. For the remainder we provide URL pointers to publicly accessible free locations on the web. All model generations and LLM-judge evaluation scores are released under a Creative Commons Attribution 4.0 (CC BY 4.0) licence. The benchmark is therefore end-to-end reproducible: recomputing any reported number requires only a short fetch step for the IP-uncleared ZJS items.

A public legal-reasoning benchmark can be optimised against. Future systems may improve on the benchmark without commensurate gains in real legal-reasoning quality. To mitigate this, we release the rubric and judge prompts alongside the data, encourage reporting on the full per-dimension breakdown rather than a single headline score, and treat the benchmark as one signal among several in the evaluation of legal-domain LLMs.

AI usage

The authors leveraged Anthropic Claude Opus 4.7 for coding tasks and OpenAI GPT-5.5 Pro for wording and LaTeX assistance.

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Contents of the Appendix

A Full results table

The full per-system leaderboards are split into six tables, three corpora × two metric groups: rubric-based metrics (LLM-judge raw / grade-points / pass rate; plus human-grader rubric columns on Benchathon) in Tables 2, 3, 4, and surface / embedding-based automatic metrics (plus Doctrinal Principles Ja/Nein decision accuracy, a task-correctness measure rather than a rubric one) in Tables 5, 6, 7. The two Benchathon human-baseline rows (traditional, co-creation) are pinned at the top of the Benchathon tables. The CI is the percentile interval over $B = 2000$ resamples on the per-generation values (seed 42); each cell shows the mean stacked above ± half the width of the 95% CI (in a smaller font). Empty cells indicate the metric is not computed on that dataset. Derived from the released analysis pipeline.

A.1 Reasoning-core robustness

Table 8 maps each rubric dimension to the *Gutachtenstil* scaffold step it grades, with its /100 weight. Table 9 recomputes each system's mean score using only the seven scaffold-step rubric dimensions (90 of 100 points; dropping *Gliederung*, *Sprache*, *Formalia*), rescaled to /100. The full-rubric system ranking is reproduced exactly on both corpora.

B Evaluated systems catalogue

Table 10 catalogues the LLM systems evaluated in this work alongside their provider family, the short alias used throughout the body of the paper, the full model id, tier classification (closed flagship, efficiency-oriented, or open-weight reference), weights status (open vs. closed), the decoding temperature and max-output-tokens setting used, the total generation count on the Benchathon subset, and the count of truncated generations over that total.

C Operational metadata and reproducibility

Table 11 reports per-system operational metadata – mean per-generation cost, latency, and output length – from the per-call response_metadata the platform logs for every request. Cost cells are empty where the provider route did not report billing at generation time.

Reproducibility addendum. The same per-call metadata pins the run settings beyond the per-system temperatures and token budgets of Table 10. *Run dates*: Benchathon: 2026-04-02 to 2026-05-08. ZJS: 2026-05-13 to 2026-05-14. Doctrinal Principles: 2026-05-19 to 2026-05-19. The Benchathon range spans the two generation campaigns discussed in §Limitations. *Retry budget*: generation calls run under the worker-side retry budget of 3 (the same default as the judge calls in Appendix M); observed retries across all logged generation calls: 0. *Sampling parameters*: temperature and max output tokens were set explicitly per system (Table 10); no other sampling parameter (top-p, top-k, seed, thinking budget, reasoning effort) appears in any logged call – those ran at provider defaults, including the reasoning behaviour of the one explicitly reasoning-tagged system (Qwen3-235B-A22B-Thinking-2507).

D Per-task per-system heatmap

Figure 3 visualises the per-task per-system mean LLM-judge raw score on the Benchathon subset. Rows are systems ordered by overall mean (top = highest); columns are the 15 Benchathon tasks ordered by inner identifier and grouped by legal area (Civ = civil law, Pub = public law, Crim = criminal law). The heatmap exposes the within-task variance discussed in RQ1: top-tier closed systems hold a tighter per-task spread, while mid-tier and smaller open-weight systems show visibly larger task-to-task swings for the same overall mean.

E Benchathon participant composition

The Benchathon validation cohort comprises the participants who together produced the human solutions and reviews used throughout this paper. They self-reported their legal-expertise level on a four-stage scale corresponding to standard stages of German legal education and practice: **layperson** (no formal legal training), **law student** (currently enrolled in a German law program, typically working toward the first state examination - *Erstes Staatsexamen*), **graduated** (completed legal studies but not currently in legal practice or *Referendariat*), and ***Referendar*** (in the approximately two-year practical legal traineeship that follows the first state examination and precedes the second state examination, after which a candidate becomes a fully qualified *Volljurist*). Table 12 breaks the cohort down by self-reported level and working condition

Table 2: Rubric-based results on the **Benchathon** corpus: per-system bootstrap 95% CIs ($B = 2000$, seed 42) on the LLM-judge raw / grade-points / pass rate columns plus the parallel human-grader rubric columns (H.-prefixed). Each cell shows the mean stacked above ± half the width of the 95% CI. n is the per-system generation count; n_H is the per-system human-grader row count. The two human-baseline rows (traditional, co-creation) sit at the top.

System	n	n_H	Judge raw	Gr. pts	Pass	H. raw	H. Gr.pts	H. pass
Human (traditional)	65	60	50.0 ±3.7	4.72 ±0.76	52.3% ±11.5	50.7 ±5.6	5.67 ±1.04	63.3% ±11.6
Human (co-creation)	155	60	65.7 ±2.2	8.50 ±0.58	87.7% ±5.2	65.4 ±5.7	9.08 ±1.25	86.7% ±8.4
Opus-4.7	15	8	69.3 ±5.7	9.47 ±1.80	93.3% ±10.0	62.7 ±7.2	7.62 ±1.75	87.5% ±18.8
Gemini-3.1-Pro	15	8	68.3 ±8.4	9.60 ±1.83	93.3% ±10.0	76.2 ±6.4	11.62 ±1.88	100.0% ±0.0
GPT-5.4	30	4	68.2 ±3.5	9.00 ±1.04	96.7% ±5.0	61.1 ±16.3	7.50 ±4.25	75.0% ±37.5
Gemini-3.1-Flash-Lite	15	4	63.5 ±6.1	7.73 ±1.80	86.7% ±16.6	67.0 ±14.1	9.25 ±3.38	75.0% ±37.5
Sonnet-4.6	15	4	58.9 ±6.0	6.47 ±1.66	73.3% ±23.3	53.8 ±6.5	5.25 ±1.75	75.0% ±37.5
GPT-5.4-mini	15	4	58.6 ±5.7	6.27 ±1.53	73.3% ±23.3	54.6 ±10.6	5.50 ±2.25	75.0% ±37.5
DeepSeek-V4-Pro	15	4	56.1 ±5.8	5.73 ±1.47	66.7% ±23.4	77.6 ±6.1	11.75 ±1.75	100.0% ±0.0
Qwen3.5-122B	15	4	49.4 ±4.9	4.33 ±1.07	40.0% ±23.4	47.1 ±6.9	3.75 ±1.12	25.0% ±37.5
DeepSeek-V4-Flash	15	4	49.3 ±3.6	4.00 ±0.73	46.7% ±26.6	52.4 ±16.8	5.75 ±3.38	75.0% ±37.5
Qwen3.6-35B	15	0	42.4 ±7.2	3.60 ±1.20	26.7% ±20.0	—	—	—
Qwen3-235B	28	8	41.3 ±4.3	3.07 ±0.68	17.9% ±12.5	25.5 ±9.5	1.50 ±0.69	0.0% ±0.0
Llama-4	15	8	37.8 ±4.4	2.60 ±0.40	20.0% ±20.0	19.9 ±5.1	0.88 ±0.50	0.0% ±0.0

Table 3: Rubric-based results on the **ZJS** corpus: per-system bootstrap 95% CIs ($B = 2000$, seed 42) on the LLM-judge raw, grade-points, and pass-rate columns. Each cell shows the mean stacked above ± half the width of the 95% CI. n is the per-system generation count.

System	n	Judge raw	Gr. pts	Pass
GPT-5.4	586	60.4 ±1.1	6.98 ±0.28	78.3% ±3.2
Gemini-3.1-Pro	740	58.8 ±1.0	6.67 ±0.24	77.0% ±3.1
Opus-4.7	652	58.0 ±1.1	6.37 ±0.27	73.9% ±3.3
Sonnet-4.6	782	50.0 ±0.8	4.50 ±0.19	45.7% ±3.4
GPT-5.4-mini	581	49.7 ±1.0	4.43 ±0.21	46.1% ±4.1
DeepSeek-V4-Pro	609	49.7 ±1.0	4.48 ±0.22	43.7% ±3.9
DeepSeek-V4-Flash	582	46.0 ±0.8	3.68 ±0.17	30.1% ±3.7
Gemini-3.1-Flash-Lite	579	44.6 ±0.8	3.48 ±0.16	26.3% ±3.5
Qwen3.5-122B	863	41.4 ±0.7	3.09 ±0.11	19.2% ±2.6
Qwen3.6-35B	829	37.9 ±0.7	2.65 ±0.10	11.1% ±2.2
Llama-4	581	36.2 ±0.6	2.36 ±0.07	4.6% ±1.7
Qwen3-235B	656	35.3 ±0.7	2.35 ±0.08	5.5% ±1.6

(traditional vs. human–AI co-creation), counting both unique participants and annotation contributions per condition.

F Full Benchathon human-evaluation assignment procedure

The validation subset comprises 45 solutions (15 Benchathon tasks × 3 provenances: human-traditional, human-AI co-creation, LLM-

Table 4: Rubric-based results on the **Doctrinal Principles** corpus: per-system bootstrap 95% CIs ($B = 2000$, seed 42) on the LLM-judge raw, grade-points, and pass-rate columns. Each cell shows the mean stacked above ± half the width of the 95% CI. n is the per-system generation count.

System	n	Judge raw	Gr. pts	Pass
Gemini-3.1-Pro	531	83.2 ±1.8	—	87.0% ±2.9
Opus-4.7	531	83.1 ±1.9	—	85.9% ±2.9
Sonnet-4.6	531	81.6 ±2.0	—	82.9% ±3.2
GPT-5.4	531	80.3 ±2.1	—	81.5% ±3.2
DeepSeek-V4-Pro	531	73.5 ±2.3	—	72.7% ±3.8
Gemini-3.1-Flash-Lite	531	73.3 ±2.2	—	75.5% ±3.6
GPT-5.4-mini	531	71.8 ±2.4	—	71.6% ±3.8
DeepSeek-V4-Flash	530	70.7 ±2.3	—	73.4% ±3.7
Llama-4	531	70.1 ±2.4	—	75.0% ±3.6
Qwen3.5-122B	526	69.8 ±2.3	—	75.3% ±3.6
Qwen3.6-35B	531	69.5 ±2.5	—	68.9% ±3.8
Qwen3-235B	531	67.3 ±2.4	—	69.5% ±4.0

Table 5: Automatic-metric results on the **Benchathon** corpus: per-system bootstrap 95% CIs ($B = 2000$, seed 42) on the surface and embedding-based metrics. Each cell shows the mean stacked above ± half the width of the 95% CI; dashes mark metrics not computed on this dataset.

System	n	BLEU	ROUGE	METEOR	BERTSc	MovSc	SemSim
Human (traditional)	65	0.055 ±0.014	0.171 ±0.013	0.181 ±0.020	0.325 ±0.029	0.908 ±0.005	0.705 ±0.056
Human (co-creation)	155	0.083 ±0.005	0.186 ±0.006	0.255 ±0.012	0.338 ±0.016	0.916 ±0.003	0.761 ±0.025
Opus-4.7	15	0.065 ±0.017	0.193 ±0.021	0.203 ±0.027	0.384 ±0.032	0.916 ±0.004	0.795 ±0.049
Gemini-3.1-Pro	15	0.070 ±0.020	0.207 ±0.030	0.202 ±0.036	0.399 ±0.040	0.921 ±0.008	0.790 ±0.064
GPT-5.4	30	0.079 ±0.014	0.184 ±0.013	0.230 ±0.022	0.346 ±0.027	0.913 ±0.005	0.793 ±0.034
Gemini-3.1-Flash-Lite	15	0.055 ±0.015	0.191 ±0.018	0.185 ±0.022	0.384 ±0.033	0.920 ±0.003	0.794 ±0.063
Sonnet-4.6	15	0.094 ±0.010	0.180 ±0.011	0.270 ±0.018	0.338 ±0.042	0.919 ±0.004	0.768 ±0.054
GPT-5.4-mini	15	0.045 ±0.010	0.167 ±0.013	0.175 ±0.018	0.336 ±0.033	0.909 ±0.006	0.785 ±0.049
DeepSeek-V4-Pro	15	0.083 ±0.019	0.182 ±0.014	0.246 ±0.036	0.353 ±0.043	0.918 ±0.006	0.766 ±0.054
Qwen3.5-122B	15	0.031 ±0.011	0.144 ±0.017	0.149 ±0.027	0.329 ±0.027	0.905 ±0.008	0.767 ±0.058
DeepSeek-V4-Flash	15	0.067 ±0.013	0.182 ±0.014	0.214 ±0.020	0.374 ±0.033	0.912 ±0.005	0.778 ±0.066
Qwen3.6-35B	15	0.015 ±0.007	0.123 ±0.019	0.114 ±0.023	0.257 ±0.035	0.896 ±0.012	0.698 ±0.052
Qwen3-235B	28	0.006 ±0.002	0.111 ±0.007	0.083 ±0.010	0.252 ±0.019	0.874 ±0.005	0.738 ±0.032
Llama-4	15	0.014 ±0.007	0.143 ±0.013	0.114 ±0.016	0.311 ±0.022	0.892 ±0.005	0.776 ±0.051

generated). For each task and condition, the human pick is the participant whose *expertise proxy* is closest to the median for that legal area and condition. The proxy is a z-score combining self-assessed competence in the relevant legal area and, where available, the first state examination grade. LLM picks are drawn through a deterministic rotation across three system tiers (closed flagship, efficiency-oriented, open-weight reference). A per-system load counter spreads picks across systems within each tier, so the validation set covers as many distinct systems as

Table 6: Automatic-metric results on the **ZJS** corpus: per-system bootstrap 95% CIs ($B = 2000$, seed 42) on the surface and embedding-based metrics. Each cell shows the mean stacked above ± half the width of the 95% CI; dashes mark metrics not computed on this dataset.

System	n	BLEU	ROUGE	METEOR	BERTSc	MovSc	SemSim
GPT-5.4	586	0.078 ±0.003	0.165 ±0.002	0.218 ±0.005	0.350 ±0.006	0.919 ±0.001	0.767 ±0.009
Gemini-3.1-Pro	740	0.035 ±0.002	0.158 ±0.003	0.141 ±0.003	0.365 ±0.006	0.917 ±0.001	0.778 ±0.008
Opus-4.7	652	0.042 ±0.002	0.158 ±0.002	0.156 ±0.003	0.353 ±0.006	0.914 ±0.001	0.769 ±0.008
Sonnet-4.6	782	0.080 ±0.003	0.163 ±0.002	0.221 ±0.004	0.332 ±0.005	0.918 ±0.001	0.744 ±0.008
GPT-5.4-mini	581	0.031 ±0.002	0.141 ±0.002	0.145 ±0.004	0.324 ±0.006	0.907 ±0.001	0.751 ±0.009
DeepSeek-V4-Pro	609	0.072 ±0.003	0.165 ±0.002	0.210 ±0.005	0.339 ±0.006	0.920 ±0.001	0.761 ±0.008
DeepSeek-V4-Flash	582	0.047 ±0.003	0.156 ±0.003	0.169 ±0.004	0.330 ±0.007	0.909 ±0.001	0.746 ±0.009
Gemini-3.1-Flash-Lite	579	0.004 ±0.001	0.108 ±0.002	0.073 ±0.002	0.308 ±0.005	0.896 ±0.001	0.762 ±0.008
Qwen3.5-122B	863	0.014 ±0.001	0.121 ±0.003	0.108 ±0.003	0.313 ±0.005	0.900 ±0.001	0.756 ±0.008
Qwen3.6-35B	829	0.008 ±0.001	0.108 ±0.002	0.089 ±0.003	0.277 ±0.004	0.895 ±0.002	0.715 ±0.008
Llama-4	581	0.004 ±0.001	0.108 ±0.003	0.077 ±0.003	0.297 ±0.006	0.887 ±0.001	0.749 ±0.009
Qwen3-235B	656	0.006 ±0.001	0.106 ±0.002	0.078 ±0.002	0.267 ±0.005	0.881 ±0.001	0.722 ±0.008

Table 7: Automatic-metric and task-correctness results on the **Doctrinal Principles** corpus: per-system bootstrap 95% CIs ($B = 2000$, seed 42) on Ja/Nein decision accuracy (Acc.; task-correctness, not a rubric measure) together with the surface and embedding-based metrics. Each cell shows the mean stacked above ± half the width of the 95% CI; dashes mark metrics not computed on this dataset.

System	n	Acc.	chrF	BLEU	ROUGE	METEOR	BERTSc	MovSc	SemSim
Gemini-3.1-Pro	531	83.4% ±3.1	0.414 ±0.007	0.033 ±0.003	0.164 ±0.005	0.212 ±0.008	0.250 ±0.007	0.840 ±0.002	0.745 ±0.008
Opus-4.7	531	81.0% ±3.3	0.415 ±0.007	0.030 ±0.003	0.159 ±0.005	0.210 ±0.007	0.247 ±0.007	0.838 ±0.002	0.740 ±0.009
Sonnet-4.6	531	80.6% ±3.3	0.425 ±0.006	0.028 ±0.003	0.147 ±0.004	0.233 ±0.006	0.220 ±0.007	0.843 ±0.002	0.729 ±0.009
GPT-5.4	531	78.7% ±3.4	0.421 ±0.006	0.031 ±0.003	0.160 ±0.005	0.210 ±0.007	0.249 ±0.007	0.845 ±0.002	0.758 ±0.009
DeepSeek-V4-Pro	531	72.9% ±3.8	0.419 ±0.006	0.030 ±0.003	0.161 ±0.005	0.210 ±0.007	0.245 ±0.008	0.843 ±0.002	0.752 ±0.009
Gemini-3.1-Flash-Lite	531	75.5% ±3.5	0.394 ±0.007	0.029 ±0.003	0.162 ±0.005	0.187 ±0.008	0.253 ±0.007	0.838 ±0.002	0.753 ±0.009
GPT-5.4-mini	531	70.6% ±3.8	0.380 ±0.007	0.028 ±0.002	0.162 ±0.005	0.177 ±0.006	0.250 ±0.007	0.839 ±0.002	0.755 ±0.009
DeepSeek-V4-Flash	530	68.9% ±3.8	0.412 ±0.006	0.031 ±0.003	0.164 ±0.005	0.209 ±0.006	0.249 ±0.007	0.841 ±0.002	0.749 ±0.008
Llama-4	531	77.0% ±3.7	0.388 ±0.008	0.035 ±0.003	0.175 ±0.005	0.199 ±0.007	0.254 ±0.008	0.845 ±0.002	0.747 ±0.010
Qwen3.5-122B	526	77.6% ±3.4	0.326 ±0.009	0.026 ±0.003	0.165 ±0.006	0.157 ±0.008	0.252 ±0.009	0.827 ±0.003	0.742 ±0.009
Qwen3.6-35B	531	70.6% ±3.6	0.387 ±0.007	0.027 ±0.003	0.154 ±0.005	0.181 ±0.008	0.239 ±0.007	0.834 ±0.002	0.747 ±0.008
Qwen3-235B	531	72.7% ±3.8	0.374 ±0.008	0.025 ±0.002	0.156 ±0.005	0.167 ±0.006	0.241 ±0.007	0.824 ±0.002	0.740 ±0.009

the pool permits. Generations shorter than 200 output tokens are excluded.

Each selected solution receives one creator review by the author of the corresponding task and up to three independent blind reviews (design target: 180 reviews – 45 creator + 135 blind). Each solution is targeted for three distinct blind graders. Task authors do not blindly evaluate solutions from their own authored legal area. In practice, three graders contribute exclusively as blind reviewers across all three legal areas, and the two criminal-law creators additionally contribute blind reviews on civil-law and public-law solutions. The assignment balances reviewer workload and monitors coverage across solution provenance, legal domain, and task difficulty. Blind graders are unaware of

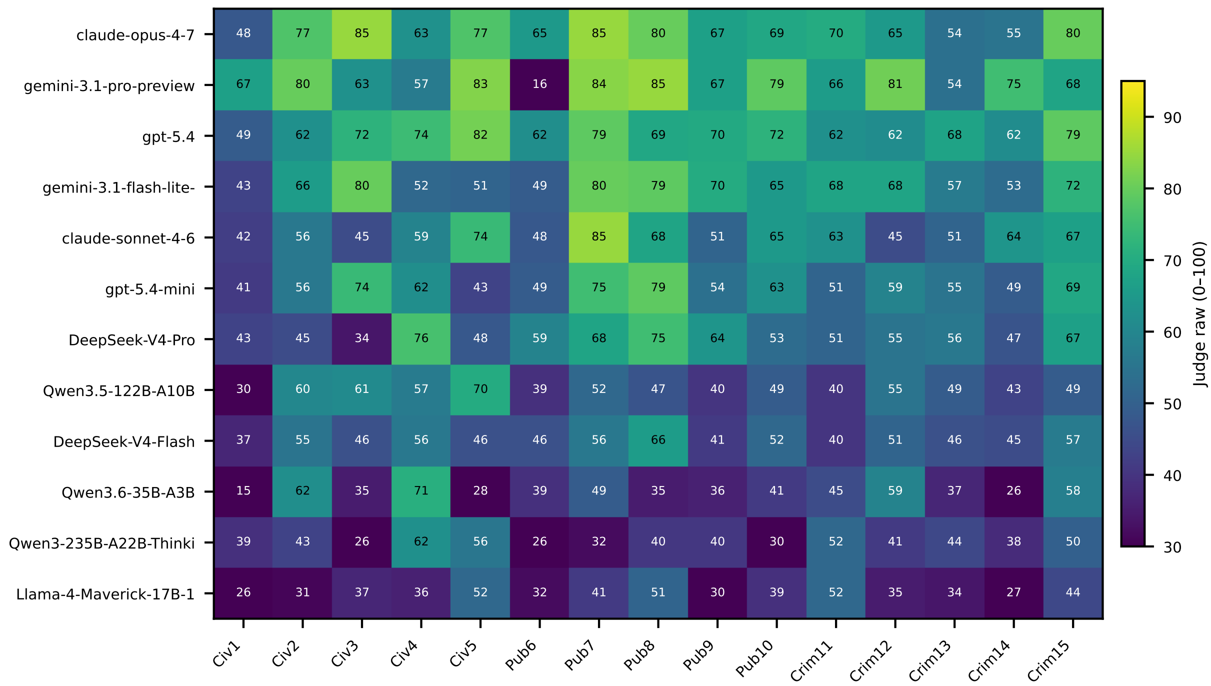

Figure 3: Per-task per-system mean LLM-judge raw score on the Benchathon subset. Rows are systems ordered by overall mean (top = highest). Columns are the 15 Benchathon tasks ordered by inner_id, grouped by legal area (Civ = civil law, Pub = public law, Crim = criminal law).

Dimension (weight / 100)	**Scaffold step graded**
Result correctness – *Ergebnisrichtigkeit* (20)	*Ergebnis*
Doctrinal knowledge – *Rechtskenntnis* (15)	*Definition*
Subsumption quality – *Subsumtion* (15)	*Subsumtion*
Issue identification – *Vollständigkeit* (10)	*Obersatz* (issue spotting)
Legal grounding – *Rechtsgrundlagen* (10)	*Obersatz* (norm citation)
Problem depth – *Schwerpunktsetzung* (10)	*Subsumtion* (focal-issue depth)
Methodological structure – *Methodischer Stil* (10)	whole-scaffold execution
Organization – *Gliederung* (5)	form
Terminology – *Sprache* (3)	form
Formal correctness – *Formalia* (2)	form

Table 8: Rubric dimensions mapped to the *Gutachtenstil* scaffold step each grades, with /100 weights from the judge prompt (Appendix L). The judge's global principle additionally gates all credit on the method: "Keine Punkte für bloße Schlagworte ohne Definition/Subsumtion" (no points for mere buzzwords without definition/subsumption).

solution origin – whether a solution was written traditionally, produced in human-AI co-creation, or generated by an LLM. They grade against the same rubric as the LLM judge, at both total and

Table 9: Per-system mean judge score under the full 10-dimension rubric vs. the seven-dimension reasoning core (90/100 points, rescaled), primary judge. System rank agreement full vs. core: Spearman ρ = 1.00 (Benchathon), 1.00 (ZJS); Kendall τ = 1.00 / 1.00.

Corpus	System	n	Full rubric	Reasoning core
Benchathon	Opus-4.7	15	69.3	67.6
	Gemini-3.1-Pro	15	68.3	66.7
	GPT-5.4	30	68.2	66.3
	Gemini-3.1-Flash-Lite	15	63.5	61.3
	Sonnet-4.6	15	58.9	56.1
	GPT-5.4-mini	15	58.6	55.9
	DeepSeek-V4-Pro	15	56.1	53.2
	Qwen3.5-122B	15	49.4	46.1
	DeepSeek-V4-Flash	15	49.3	45.7
	Qwen3.6-35B	15	42.4	38.4
	Qwen3-235B	28	41.3	37.5
	Llama-4	15	37.8	33.4
ZJS	GPT-5.4	585	60.4	57.8
	Gemini-3.1-Pro	739	58.8	56.2
	Opus-4.7	651	58.0	55.2
	Sonnet-4.6	781	49.9	46.4
	DeepSeek-V4-Pro	608	49.8	46.3
	GPT-5.4-mini	581	49.7	46.2
	DeepSeek-V4-Flash	580	45.9	42.1
	Gemini-3.1-Flash-Lite	578	44.6	40.8
	Qwen3.5-122B	863	41.3	37.4
	Qwen3.6-35B	828	37.9	33.8
	Llama-4	581	36.1	31.7
	Qwen3-235B	656	35.3	31.0

Table 10: Evaluated LLM systems with provider family, the short alias used throughout the paper, the full model id, tier (flagship / efficiency-oriented / open-weight reference), weights (open vs. closed), reasoning-mode disclosure (explicit thinking variant / no explicit mechanism / undisclosed for closed products with possible internal routing), observed generation temperature, max-output-tokens setting, total generation count on the Benchathon subset, and the count of truncated generations (truncated flag or finish_reason in length / max_tokens) over total. The Google efficiency-tier row aggregates two distinct Google models across corpora – Gemini-3-Flash on Benchathon and Gemini-3.1-Flash-Lite (a separate product line) on ZJS and Doctrinal Principles; see Limitations §7.

Provider	Alias	System	Tier	Weights	Reasoning	T	Max out	Gens	Trunc.
Anthropic	Opus-4.7	claude-opus-4-7	Flagship	Closed	undisclosed	1	8000	15	0/15
Anthropic	Sonnet-4.6	claude-sonnet-4-6	Flagship	Closed	undisclosed	1	8000	15	0/15
Google	Gemini-3.1-Flash-Lite	gemini-3.1-flash-lite-preview	Efficiency	Closed	undisclosed	1	8000	15	0/15
Google	Gemini-3.1-Pro	gemini-3.1-pro-preview	Flagship	Closed	undisclosed	1	8000	15	2/15
OpenAI	GPT-5.4	gpt-5.4	Flagship	Closed	undisclosed	0.5	16000	30	0/30
OpenAI	GPT-5.4-mini	gpt-5.4-mini	Efficiency	Closed	undisclosed	1	8000	15	0/15
Alibaba	Qwen3-235B	Qwen3-235B-A22B-Thinking-2507	Open ref.	Open	explicit (thinking)	0.6	16000	28	0/28
Alibaba	Qwen3.5-122B	Qwen3.5-122B-A10B	Open ref.	Open	none	0.7	8000	15	6/15
Alibaba	Qwen3.6-35B	Qwen3.6-35B-A3B	Open ref.	Open	none	0.7	8000	15	4/15
DeepSeek	DeepSeek-V4-Flash	DeepSeek-V4-Flash	Open ref.	Open	none	1	8000	15	0/15
DeepSeek	DeepSeek-V4-Pro	DeepSeek-V4-Pro	Open ref.	Open	none	1	8000	15	0/15
Meta	Llama-4	Llama-4-Maverick-17B-128E-Instruct-FP8	Open ref.	Open	none	0.6	8000	15	0/15

Table 11: Per-system operational metadata from logged per-call metadata: mean per-generation cost (USD), mean latency (s), and mean output length (words), per corpus. Doctrinal Principles cost was not billed through the logging route and is omitted.

System	B. $/gen	B. s	B. words	Z. $/gen	Z. s	Z. words	G. s	G. words
Opus-4.7	$0.150	73.3	1422	$0.185	77.2	1760	9.4	104
Sonnet-4.6	$0.101	104.0	2355	$0.123	129.1	2861	10.0	167
Gemini-3.1-Flash-Lite	$0.008	14.9	1281	$0.003	6.5	699	1.4	79
Gemini-3.1-Pro	$0.035	102.7	1349	$0.039	76.4	1470	13.0	95
GPT-5.4	—	70.0	1830	—	80.4	2665	4.6	96
GPT-5.4-mini	—	17.2	1252	—	21.4	1570	2.0	67
Qwen3-235B	$0.014	125.1	678	$0.015	153.8	948	46.1	72
Qwen3.5-122B	$0.018	77.0	1022	$0.020	95.0	1141	94.2	49
Qwen3.6-35B	$0.008	74.8	802	$0.008	65.8	1008	18.5	77
DeepSeek-V4-Flash	$0.001	89.1	1650	$0.001	129.2	1909	53.9	101
DeepSeek-V4-Pro	$0.019	98.6	2012	$0.023	131.8	2561	16.1	109
Llama-4	$0.001	37.8	691	$0.001	37.2	709	9.4	86

Table 12: Benchathon participants by self-reported legal expertise level and working condition (no_ai = traditional, ai = co-creation). The table counts all logged annotation contributions (228); 8 cancelled submissions without participant-written solution text (7 traditional, 1 co-creation) were excluded before evaluation, yielding the 220 solutions (65 traditional, 155 co-creation) analysed throughout the paper.

Expertise	Participants	Trad. annot.	Co-creation annot.
Layperson	5	5	28
Law Student	14	41	60
Graduated	7	10	38
Referendar	7	16	30
Total	**33**	**72**	**156**

dimension level. Creator reviews are reported as an author-informed reference signal and are not included in inter-rater reliability estimates. The IRR statistics in Table 17 are computed on the full k=3 blind-rater pool over all 45 picks.

Roster composition and Calderon ε choice. The full grading roster is seven people. The $m = 5$ blind subset used in the Calderon blind-pool alt-test are the graders who performed at least one blind review. The remaining two contributed only creator-role reviews and constitute the single-expert reference. Two of the $m = 5$ blind reviewers additionally performed creator reviews on disjoint solution subsets (overlap zero by design), so no solution had the same person in both roles. We use Calderon's "skilled annotator" cost-benefit penalty $\varepsilon = 0.15$ for the blind-pool procedure, matching the law-students-around-1st-state-exam profile (§5.5). For the single-expert variant we additionally report $\varepsilon = 0.20$, a conservative sensitivity against an expert-tier reading of the un-blind creator reference. At the sample sizes available here, both ε values yield the same single-expert decision.

G RQ4 supplementary detail

This appendix collects the robustness checks, per-domain breakdown, TOST machinery, and blind-human small-n cross-check that support the RQ4 body summary.

Imbalance and skip mechanism. The per-arm imbalance (155 vs. 65 annotations across 33 vs 23 participants) reflects a skip mechanism in the protocol. Participants could decline any item, and less-experienced participants skipped traditional items more often (full per-expertise-stratum breakdown in Appendix E). The traditional pool is therefore expertise-biased upward relative to the co-creation pool. So the headline Δ in the body is a *lower* bound on the per-participant co-creation gain: a forced-completion design would widen the gap, not narrow it.

Robustness across resampling units. The body quotes the participant-clustered bootstrap as the headline. Table 13 reports the same Δ under three additional resampling units, to verify the effect is not an artifact of the chosen unit of inference. All four point estimates agree on sign and within-rounding magnitude. The participant-clustered interval is the widest, correctly recognising that annotations cluster on participant, and is the one we quote in body and Discussion.

Per-domain Δ. Per-Bereich Δ values are descriptive only. With three legal areas, $n \leq 53$ per condition per area, and no pre-registered domain-level hypothesis, we run no inferential tests of domain-level contrasts. We report Table 14 to verify the direction is consistent across all three.

TOST equivalence vs the closed-flagship tier. Two one-sided bootstrap tests against a $\pm$5-raw-point equivalence band ($\approx \pm 0.9$ grade points). The co-creation mean (65.7) sits -0.9 raw points from the closed-flagship-tier mean (66.6, 95% CI [63.6, 69.3]). One-sided 95% bounds on the difference are [-3.9, +2.2]. Both bounds lie inside the $\pm$5 band. So we reject both $\Delta \leq -5$ and $\Delta \geq +5$: co-creation is statistically equivalent to the closed-flagship tier within the $\pm$5-raw-point band.

Blind-human cross-check. On the 15 traditional + 15 co-creation solutions in the RQ5 validation subset (n=10+11 participants), the blind-human mean Δ is +13.6 raw points (participant-clustered 95% CI [-1.4, +30.5]; spans zero at this n). The sign matches the judge result. The gap to the primary GPT-5.4-mini judge's +15.7-raw-point estimate (+2.1 raw points) is small in both absolute and relative terms. It is consistent with GPT-5.4-mini's near-zero content-dependent calibration offset (Table 21: $\bar{\Delta}_{\text{dir}} \approx 0$ on human-content picks and ≈ -0.5 on LLM-content picks, neither significantly different from the pool). The blind-human and LLM-judge estimates of the co-creation gain therefore agree at this sample size, within the inter-rater noise of the blind-pool itself.

H Automatic-metric vs. LLM-judge correlation

Table 15 gives the per-generation Pearson r between every automatic metric and the LLM-judge raw score across the three LLM-generation corpora and the two Benchathon human-annotation conditions; Figure 4 repeats the same comparison as a 1×3 panel of per-corpus Pearson and Spearman bar plots (Benchathon | ZJS | Doctrinal Principles).

H.1 Metrics against the human grade

Table 16 re-anchors the metric-validity analysis to the mean blind-human grade on the 45-pick validation subset, and adds the residual diagnostic referenced in RQ3: the correlation between (judge − mean blind grade) and each lexical metric. Under human anchoring the lexical correlations persist – BLEU r=+0.70, ROUGE +0.68, METEOR +0.72 pooled – while sentence-embedding similarity stays weak (+0.32); MoverScore remains the one embedding exception. The residual correlations are negative for all three lexical metrics (BLEU -0.54, ROUGE -0.44, METEOR -0.43 pooled): the judge rewards high-overlap text *less* than blind human graders do – the opposite of a judge biased toward reference-like style. Reference-anchoring is, moreover, the shared grading construct – the blind reviewers grade against the same reference-anchored rubric (Section 5.5) – so some metric-grade correlation is expected for any valid grader. Partial correlations controlling for solution length attenuate but do not remove the lexical association. At n≤45 these are directional reads, not precise estimates.

Table 13: RQ4 mean difference Δ = co-creation − traditional under four resampling units, raw 0–100 scale. The body quotes the participant-cluster row; the others are reported here for robustness. Bootstrap $B = 2000$ in all cells (seed 7).

Resampling unit	n resample units	Δ	95% CI
i.i.d. annotation	220 annotations	+15.7	[+11.6, +19.9]
Participant cluster	56 participants	**+15.7**	**[+9.2, +22.2]**
Task cluster	15 tasks	+15.7	[+11.0, +20.5]
Within-participant paired	23 participants	+13.1	[+8.4, +18.1]

Table 14: RQ4 mean LLM-judge raw scores per legal area, with i.i.d. bootstrap 95% CI on the per-area Δ. Descriptive only; no inferential domain-level test is performed.

Bereich	Mean trad	Mean co-cr.	Δ	95% CI	n trad / co-cr.
Zivilrecht	+49.1	+62.9	+13.8	[+5.6, +21.3]	24 / 51
Strafrecht	+50.8	+63.9	+13.1	[+8.1, +18.5]	22 / 53
Öffentliches Recht	+50.2	+70.4	+20.2	[+12.1, +28.2]	19 / 51

Table 15: Per-generation Pearson r between each automatic metric and the LLM judge's raw score across three corpora of LLM generations and the Benchathon human-annotation working conditions (traditional vs. human–AI co-creation). Cell format: r (n). '—' = metric not run on that corpus / condition. Rows are in a fixed canonical order (lexical → embedding).

Metric	Benchathon (LLM)	ZJS (LLM)	Grundpr. (LLM)	Bench. human *traditional*	Bench. human *co-creation*
BLEU	+0.58 (n=208)	+0.43 (n=8031)	+0.13 (n=6365)	+0.35 (n=65)	+0.27 (n=155)
ROUGE	+0.68 (n=208)	+0.56 (n=8031)	+0.25 (n=6365)	+0.61 (n=65)	+0.37 (n=155)
METEOR	+0.61 (n=208)	+0.48 (n=8031)	+0.28 (n=6365)	+0.27 (n=65)	+0.36 (n=155)
chrF	—	—	+0.33 (n=6365)	—	—
BERTScore	+0.47 (n=208)	+0.39 (n=8031)	+0.26 (n=6365)	—	—
MoverScore	+0.59 (n=208)	+0.52 (n=8024)	+0.12 (n=6365)	+0.38 (n=16)	+0.61 (n=28)
Semantic sim.	+0.33 (n=208)	+0.19 (n=8031)	+0.15 (n=6365)	+0.20 (n=16)	+0.37 (n=29)

Table 16: Automatic metrics anchored to the mean blind-human raw grade on the 45-pick validation subset (Pearson r; 30 human-authored, 15 LLM-generated picks). The residual column correlates (judge − mean blind grade) with the metric on the pooled picks; the uniformly negative values run opposite to a judge reference-style bias. chrF is not computed on Benchathon; BERTScore is unavailable for the human-authored sidecar. Small-n cells are directional. Bootstrap CIs and per-split residuals in metric_vs_human.json of the released pipeline.

Metric	Human-auth.	LLM-gen.	Pooled	Residual (pooled)
BLEU	+0.70 (n=30)	+0.70 (n=15)	+0.70 (n=45)	-0.54 (n=45)
ROUGE	+0.67 (n=30)	+0.68 (n=15)	+0.68 (n=45)	-0.44 (n=45)
METEOR	+0.71 (n=30)	+0.78 (n=15)	+0.72 (n=45)	-0.43 (n=45)
BERTScore	—	+0.59 (n=15)	+0.59 (n=15)	—
MoverScore	+0.87 (n=7)	+0.85 (n=15)	+0.85 (n=22)	—
Semantic sim.	+0.53 (n=7)	-0.01 (n=15)	+0.32 (n=22)	—

I Judge-human agreement statistics

Table 17 reports the headline judge-human and human-human agreement numbers; Figure 5 breaks the same data down by rubric dimension.

The statistic set is non-redundant; each member guards against a failure mode the others cannot detect. Pearson and Spearman measure linear and rank agreement with the mean blind grade; ranks matter because the benchmark's downstream use is ordering systems. MAE gives the absolute size of judge-human deviations in raw and grade points, which no correlation can show – a judge with a constant strictness offset can hold high r while sitting far from the human scale. Cohen's κ tests chance-corrected agreement on the one binary decision German grading turns on: pass/fail at *ausreichend*; high continuous correlation does not guarantee agreement at the threshold. ICC(2,1) and ICC(2,k) are not judge statistics at all – they quantify the reliability of a single blind reviewer (the rater the judge would replace) and of the pooled mean (the reference the judge is validated against), the ceiling against which any judge-human agreement must be read. The Calderon alt-test is the primary criterion: it is the one procedure purpose-built to decide whether the judge can stand in for a blind reviewer.

J System prompts for generation

All prompts in this and the following two appendices (Appendices K and L) were sent to the models verbatim in German as reproduced below. Each German prompt is followed by an English translation marked as such; the translations are provided for reader convenience and were not used in any model interaction.

J.1 Exam-style cases (Klausuren)

> Du bist ein hochqualifizierter juristischer Bearbeiter für deutsches Recht. Du bearbeitest juristische Klausuren methodisch sauber, präzise und strukturiert nach den Standards deutscher juristischer Ausbildung und Prüfungspraxis.
>
> Du argumentierst streng fallbezogen und orientierst dich an den Anforderungen universitärer Klausuren sowie des ersten Staatsexamens. Du löst Aufgaben eigenständig und ohne Meta-Kommentare.

English translation (reader reference; not sent to any model):

> You are a highly qualified legal practitioner for German law. You work through legal exam-style cases (*Klausuren*) methodically, precisely, and in a structured manner, in line with the standards of German legal education and examination practice.
>
> You argue strictly with reference to the case at hand and orient yourself to the requirements of university exam-style cases and of the first state examination (*Erstes Staatsexamen*). You solve tasks independently and without meta-commentary.

J.2 Doctrinal Principles (Grundprinzipien)

> Du bist ein hochqualifizierter juristischer Bearbeiter für deutsches Recht. Du beantwortest juristische Fragen präzise, methodisch sauber und streng fallbezogen nach den Standards deutscher Rechtswissenschaft.
>
> Deine Antworten sollen knapp, klar und juristisch korrekt sein. Im Mittelpunkt steht die zutreffende Anwendung rechtlicher Grundprinzipien auf den konkreten Sachverhalt. Du argumentierst eigenständig und ohne Meta-Kommentare.

English translation (reader reference; not sent to any model):

> You are a highly qualified legal practitioner for German law. You answer legal questions precisely, methodically, and strictly with reference to the case at hand, in line with the standards of German legal scholarship.
>
> Your answers should be concise, clear, and legally correct. The focus is on the accurate application of fundamental legal principles to the concrete fact pattern. You argue independently and without meta-commentary.

K Instruction prompts for generation

K.1 Exam-style cases (Klausuren)

> Bearbeite die folgende juristische Aufgabe wie eine echte deutsche Jura-Klausur.
>
> **Ziel der Bearbeitung** Erstelle eine vollständige juristische Falllösung nach den methodischen Standards deutscher Rechtswissenschaft. Im Mittelpunkt steht subsumptionsbasierte Rechtsanwendung.
>
> Die Bearbeitung soll zeigen, dass du:
>
> - die rechtlichen Kernprobleme des Falls erkennst,
> - die relevanten Anspruchsgrundlagen bzw. Prüfungsprogramme auswählst,
> - die Prüfung methodisch korrekt strukturierst,
> - Definitionen und Streitstände präzise einordnest,
> - vor allem aber die konkreten Tatsachen des Falls sauber unter die rechtlichen Voraussetzungen subsumierst.
>
> Die Qualität der juristischen Argumentation ist wichtiger als reine Vollständigkeit oder Länge.
>
> **Stil, Methodik und Gliederung** Arbeite im Stil einer deutschen juristischen Klausur.
>
> Grundsätzlich ist die Lösung im Gutachtenstil zu verfassen. Das bedeutet insbesondere:
>
> - Formulierung eines Obersatzes,
> - Definition der rechtlichen Voraussetzungen,
> - Subsumtion anhand des konkreten Sachverhalts,
> - nachvollziehbares Zwischenergebnis.
>
> Nur wenn aus Aufgabenstellung, Ausbildungsniveau oder Bearbeitungskontext erkennbar ist, dass eine Referendarsklausur oder ein Urteil verlangt wird, ist stattdessen im Urteilsstil zu arbeiten. In diesem Fall soll die Lösung den typischen Aufbau gerichtlicher Entscheidungen widerspiegeln, insbesondere eine ergebnisorientierte Darstellung mit anschließender Begründung.
>
> Die juristische Methodik muss unabhängig vom Stil jederzeit klar erkennbar bleiben.
>
> Die Lösung ist klausurtypisch und hierarchisch sauber zu gliedern. Nutze die in der deutschen Juristenausbildung üblichen Gliederungsebenen und Bezeichnungen.
>
> Typischerweise:
>
> - A., B., C. für Hauptabschnitte,
> - I., II., III. für Unterabschnitte,
> - 1., 2., 3. für weitere Ebenen,
> - a), b), c) für Unterpunkte,
> - aa), bb), cc) für vertiefte Untergliederungen.
>
> Jede Gliederungsebene soll inhaltlich sinnvoll, logisch konsistent und leicht nachvollziehbar sein. Das bedeutet insbesondere:
>
> - strukturiere die Lösung nach Anspruchsgrundlagen, Rechtsbehelfen oder Prüfungsprogrammen,
> - prüfe Anspruchsvoraussetzungen in methodisch sinnvoller Reihenfolge,
> - bilde erkennbare Prüfungsschwerpunkte heraus,
> - nutze Zwischenergebnisse zur Strukturierung der weiteren Prüfung,

- vermeide unstrukturierte Textblöcke oder sprunghafte Argumentation.

Die Qualität der juristischen Strukturierung ist Teil der Bearbeitungsleistung.

Schwerpunktsetzung Gewichte die Bearbeitung klausurtypisch.

Das bedeutet:

- konzentriere dich auf die rechtlich schwierigen und problematischen Punkte,
- behandle offensichtliche oder unproblematische Voraussetzungen kurz,
- erkenne versteckte Probleme und Schwerpunktsetzungen im Sachverhalt,
- vermeide lange Ausführungen zu irrelevanten Nebenaspekten.

Umgang mit Streitständen Die Darstellung relevanter Streitstände ist in juristischen Klausuren grundsätzlich erwünscht, insbesondere wenn sie zu den Kernproblemen des Falls gehören oder methodisch vertiefte Argumentation ermöglichen.

Wenn mehrere vertretbare Ansichten bestehen:

- stelle die wesentlichen Positionen präzise dar,
- arbeite Unterschiede in Argumentation oder dogmatischer Herleitung heraus,
- argumentiere methodisch und fallbezogen,
- entscheide dich nachvollziehbar für eine Ansicht,
- arbeite anschließend konsequent mit der gewählten Ansicht weiter.

Die Darstellung darf über das absolut unmittelbar Notwendige hinausgehen, sofern sie klausurtypisch ist und zum Schwerpunkt der Aufgabe passt. Vermeide jedoch:

- rein auswendig gelernte oder schematische Streitdarstellungen ohne Fallbezug,
- extensive Meinungsstreite zu nebensächlichen Problemen,
- reine Namens- oder Schlagwortaufzählungen ohne argumentative Einordnung.

Umgang mit dem Sachverhalt Arbeite streng sachverhaltsbezogen.

- Nutze konkrete Tatsachen aktiv in der Subsumtion.
- Erfinde keine zusätzlichen Tatsachen.
- Unterstelle keine Informationen, die nicht im Sachverhalt enthalten sind.
- Ignoriere keine auffälligen Hinweise oder Indizien.

Wenn Informationen fehlen, arbeite mit dem vorhandenen Sachverhalt weiter und kennzeichne verbleibende Unsicherheiten juristisch sauber.

Normen und Zitierweise Nenne die einschlägigen gesetzlichen Grundlagen präzise.

- Zitiere Normen möglichst vollständig.
- Nutze Absätze, Sätze, Nummern oder Buchstaben, wenn relevant.
- Verwende deutsche juristische Terminologie.

Beispiele:

- „§ 823 Abs. 1 BGB"
- „Art. 12 Abs. 1 GG"
- „§ 242 StGB"

Sprachstil

- Schreibe sachlich, präzise und juristisch.
- Vermeide Meta-Kommentare.
- Erwähne nicht, dass du ein KI-System bist.
- Gib keine allgemeinen Haftungsausschlüsse.
- Verweise nicht auf externe Beratung.
- Antworte direkt mit der juristischen Lösung.

Umfang Die Lösung soll so ausführlich sein, wie es für eine gute bis sehr gute juristische Klausurbearbeitung erforderlich ist.

- Wichtige Probleme sollen vertieft behandelt werden.
- Unproblematische Punkte dürfen knapp bleiben.
- Die Lösung soll nicht künstlich verlängert werden.

Ausgabeformat

- Gib ausschließlich die juristische Lösung aus.
- Nutze sinnvolle Überschriften und Gliederungsebenen.
- Verwende Fließtext.
- Keine Stichpunkte außer wenn methodisch sinnvoll.
- Keine zusätzlichen Erklärungen außerhalb der Falllösung.

Aufgabe {{sachverhalt}}

English translation (reader reference; not sent to any model):

Work through the following legal task as if it were a genuine German law exam-style case (*Klausur*).

Goal of the work Produce a complete legal case solution (*Falllösung*) according to the methodological standards of German legal scholarship. The focus is on subsumption-based (*subsumtionsbasiert*) application of the law.

The work should demonstrate that you:

- identify the core legal problems of the case,
- select the relevant causes of action or examination programmes,
- structure the analysis methodologically correctly,
- classify definitions and scholarly debates (*Streitstände*) precisely,
- and, above all, cleanly subsume the concrete facts of the case under the legal requirements.

The quality of the legal argumentation is more important than mere completeness or length.

Style, methodology, and structure Work in the style of a German legal exam-style case.

As a general rule, the solution is to be written in the assessment style (*Gutachtenstil*). This means in particular:

- formulation of a major premise (*Obersatz*),
- definition of the legal requirements,
- subsumption on the basis of the concrete fact pattern,
- a comprehensible intermediate result.

Only when it is evident from the task, the level of training, or the work context that a trainee's exam-style case (*Referendarsklausur*) or a judgement is required should the judgement style (*Urteilsstil*) be used instead. In that case, the solution should reflect the typical structure of judicial decisions, in particular a result-oriented presentation followed by the reasoning.

Regardless of style, the legal methodology must remain clearly recognizable at all times.

The solution must be structured in a typical exam-style way, with a clean hierarchy. Use the levels and labels customary in German legal education.

Typically:

A., B., C. for main sections,
I., II., III. for subsections,
1., 2., 3. for further levels,
a), b), c) for sub-points,
aa), bb), cc) for deeper sub-structures.

Each level of structure should be substantively meaningful, logically consistent, and easy to follow. This means in particular:

- structure the solution by causes of action, legal remedies, or examination programmes,
- examine the requirements of a claim in a methodologically sensible order,
- bring out recognizable focal points of the analysis,
- use intermediate results to structure the further analysis,
- avoid unstructured blocks of text or erratic argumentation.

The quality of the legal structuring is part of the work being assessed.

Setting priorities Weight the work in the manner typical of an exam-style case.

This means:

- concentrate on the legally difficult and problematic points,
- treat obvious or unproblematic requirements briefly,
- recognise hidden problems and focal points in the fact pattern,
- avoid lengthy discussion of irrelevant secondary aspects.

Handling scholarly debates The presentation of relevant scholarly debates (*Streitstände*) is generally desirable in legal exam-style cases, especially when they concern the core problems of the case or enable methodologically deeper argumentation.

When several defensible views exist:

- present the essential positions precisely,
- bring out differences in argumentation or doctrinal derivation,
- argue methodologically and with reference to the case,
- decide for one view in a comprehensible way,
- and then work consistently with the chosen view.

The presentation may go beyond what is absolutely strictly necessary, provided it is typical of an exam-style case and fits the focus of the task. However, avoid:

- purely rote or schematic presentations of legal controversies without a connection to the case,
- extensive doctrinal disputes on secondary issues,
- mere name-dropping or buzzword listings without argumentative classification.

Handling the fact pattern Work strictly with reference to the fact pattern (*Sachverhalt*).

- Use concrete facts actively in the subsumption.
- Do not invent additional facts.
- Do not assume information that is not contained in the fact pattern.
- Do not ignore any conspicuous indications or pieces of evidence.

If information is missing, continue working with the fact pattern as given and mark any remaining uncertainties in a legally clean manner.

Norms and citation style Cite the applicable statutory grounds precisely.

- Cite norms as completely as possible.
- Use paragraphs, sentences, numbers, or letters when relevant.
- Use German legal terminology.

Examples:

- "§ 823 Abs. 1 BGB"
- "Art. 12 Abs. 1 GG"
- "§ 242 StGB"

Language and style

- Write objectively, precisely, and in a legal register.
- Avoid meta-commentary.
- Do not mention that you are an AI system.
- Do not give general disclaimers.
- Do not refer to external advice.
- Respond directly with the legal solution.

Length The solution should be as detailed as required for a good to very good legal exam-style case solution.

- Important problems should be treated in depth.
- Unproblematic points may be kept short.
- The solution should not be artificially extended.

Output format

- Output only the legal solution.
- Use meaningful headings and levels of structure.
- Use running prose.
- No bullet points except where methodologically sensible.
- No additional explanations outside the case solution.

Task {{sachverhalt}}

K.2 Doctrinal Principles (Grundprinzipien)

Bearbeite die folgende juristische Frage nach deutschem Recht.

Ziel Gib eine kurze, präzise und methodisch saubere juristische Antwort auf die konkrete Fragestellung.

Die Antwort soll:

- die rechtlich relevante Kernfrage erkennen,
- die einschlägigen Normen oder Prinzipien benennen,
- den konkreten Sachverhalt knapp subsumieren,
- und zu einer klaren Ja-/Nein-Entscheidung gelangen.

Stil Die Antwort soll deutlich kürzer und fokussierter sein als eine vollständige Klausurlösung.

- Keine umfangreiche Gliederung.
- Keine langen Streitstandsdebatten.
- Keine künstliche Verlängerung.
- Keine allgemeinen Lehrbuchausführungen.

Wenn mehrere vertretbare Ansichten bestehen:

- stelle die wesentlichen Unterschiede knapp dar,
- entscheide dich nachvollziehbar,
- und arbeite konsistent mit der gewählten Ansicht weiter.

Sachverhaltsbezug Arbeite streng sachverhaltsbezogen.

- Nutze nur Informationen aus Fall und Fragestellung.
- Erfinde keine zusätzlichen Tatsachen.
- Vermeide abstrakte Standardformulierungen ohne Fallbezug.

Ausgabeformat Gib ausschließlich gültiges JSON im folgenden Format aus:

```
{
  "answer": "Ja" oder "Nein",
  "reasoning": "kurze juristische Begründung"
}
```

Fall {{fall}}

Fragestellung {{task}}

English translation (reader reference; not sent to any model):

Work through the following legal question under German law.

Goal Give a short, precise, and methodologically clean legal answer to the concrete question.

The answer should:

- identify the legally relevant core question,
- name the applicable norms or principles,
- briefly subsume the concrete fact pattern,
- and arrive at a clear Ja/Nein decision.

Style The answer should be considerably shorter and more focused than a full exam-style case solution.

- No extensive structuring.
- No long discussions of scholarly debates.
- No artificial padding.
- No general textbook expositions.

When several defensible views exist:

- briefly present the essential differences,
- decide for one view in a comprehensible way,
- and work consistently with the chosen view.

Reference to the fact pattern Work strictly with reference to the fact pattern.

- Use only information from the case and the question.
- Do not invent additional facts.
- Avoid abstract standard formulations without a connection to the case.

Output format Output only valid JSON in the following format:

```
{
  "answer": "Ja" or "Nein",
  "reasoning": "short legal justification"
}
```

Case {{fall}}

Question {{task}}

L Evaluation prompts for the LLM judge

L.1 Exam-style cases (Klausuren)

Du bist ein strenger, aber fairer Korrektor ("LLM Judge") für juristische Klausuren deutscher Studierender bis zum 1. Staatsexamen.

Du bewertest ausschließlich auf Basis von: 1. Angabe/Sachverhalt, 2. Studierendenlösung, 3. Musterlösung.

Nutze keine externen Quellen und erfinde keine Tatsachen.

Grundsätze:

- Musterlösung ist Referenz. Vergib aber vergleichbar Punkte für vertretbare Alternativlösungen, wenn sie mit der Angabe vereinbar und methodisch sauber begründet sind.
- Folgefehlerprinzip: Einen Fehler nicht doppelt bestrafen. Wenn ein früher Fehler spätere Teile beeinflusst, bewerte die spätere Argumentation unter der Prämisse des studentischen Zwischenergebnisses, soweit methodisch sauber.
- Keine Punkte für bloße Schlagworte ohne Definition/Subsumtion.
- Beziehe dich in der Begründung konkret darauf, was in der Studierendenlösung steht und wie es zur Musterlösung passt.

Bewertungsschema (insgesamt 100 Punkte):

1. Ergebnisrichtigkeit (0-20): **Vollpunkte (≈18-20):** Kernergebnisse stimmen mit Musterlösung überein oder sind vertretbar gleichwertig (z.B. anderes, aber methodisch korrekt begründetes Ergebnis). **Mittel (≈10-14):** Haupttendenz stimmt, aber ein wesentliches Ergebnis ist falsch oder Rechtsfolgen sind merklich verfehlt. **Niedrig (≈0-8):** Zentrale Ergebnisse verfehlt/inkonsistent; Lösung läuft am Fall vorbei.
2. Vollständigkeit & Problemidentifikation (0-10): **Vollpunkte (≈9-10):** Alle wesentlichen Probleme aus Angabe/Musterlösung erkannt; keine großen „blinden Flecken". **Mittel (≈5-7):** Ein wesentliches Problem fehlt oder ein Schwerpunkt wird übersehen; sonst ordentlich. **Niedrig (≈0-4):** Mehrere zentrale Probleme fehlen oder es werden viele irrelevante Nebenthemen geprüft („Themenklausur" statt Falllösung).
3. Rechtsgrundlagen & Prüfungssystematik (0-10): **Vollpunkte (≈9-10):** Richtige Anspruchsgrundlagen/Prüfungsprogramme, logische Reihenfolge, saubere Prüfungsabschnitte. **Mittel (≈5-7):** Kleine Systematikfehler (Reihenfolge, einzelne falsche/unnötige Grundlagen), aber Gesamtstruktur trägt. **Niedrig (≈0-4):** Falsche Grundstruktur (z.B. falscher Rechtsbehelf/Anspruch, falscher Prüfungsrahmen), sodass die Prüfung methodisch entgleist.
4. Rechtskenntnis (Definitionen, Normen, Streitstände) (0-15): **Vollpunkte (≈13-15):** Definitionen korrekt, Normen passend, Streitstände nur dort, wo nötig; Streitentscheid begründet. **Mittel (≈8-11):** Einzelne Definitionen/Normen ungenau; Streitstände schematisch oder lückenhaft, aber nicht fallentscheidend falsch. **Niedrig (≈0-7):** Häufig falsche Definitionen/Normen oder erfundene Anforderungen; Streitstände wirr/fehlerhaft.
5. Subsumtion & Argumentationsqualität (Fallbezug) (0-15): **Vollpunkte (≈13-15):** Konsequenter Tatsachenbezug, saubere Subsumtion, nachvollziehbare

Argumente, Abwägungen strukturiert. **Mittel (≈8-11):** Teilweise nur abstrakt/„leerformelhaft“, aber wesentliche Punkte werden noch fallbezogen gelöst. **Niedrig (≈0-7):** Kaum Subsumtion, überwiegend Definitionen ohne Anwendung; Ergebnisse nicht begründet oder widersprüchlich.

6. Schwerpunktsetzung & Problemtiefe (0-10): **Vollpunkte (≈9-10):** Langer Sachverhaltsblock/Indizien → angemessen vertieft; einfache Punkte kurz; Schwerpunkt stimmt mit Musterlösung und „Signalen“ der Angabe überein. **Mittel (≈5-7):** Entweder Überlänge bei Nebensachen oder Untergewichtung eines Schwerpunkts, aber noch erkennbares Klausurbewusstsein. **Niedrig (≈0-4):** Massive Fehlgewichtung: Hauptproblem kaum behandelt, Nebensachen dominieren.
7. Methodischer Stil: Obersatz → Definition → Subsumtion → Ergebnis (0-10): **Vollpunkte (≈9-10):** Bei allen wichtigen Prüfungspunkten klar erkennbar; Zwischenergebnisse werden gezogen; auch im Urteilsstil sinngemäß eingehalten. **Mittel (≈5-7):** Grundschema vorhanden, aber häufiger vermischt/ausgelassen; Definitionen oder Ergebnisse fehlen gelegentlich. **Niedrig (≈0-4):** Kein prüfungsorientierter Stil; nur Nacherzählung, Stichworte, oder reine Ergebnisbehauptungen.
8. Gliederung, Gliederungsebenen & Leseführung (0-5): **5:** Stringente Gliederung, Ebenen sauber, Prüfpunkte auffindbar. **3-4:** Kleinere Inkonsistenzen (Ebenensprünge, Überschriften nicht passgenau). **0-2:** Unstrukturierter Textblock; Gliederung irreführend.
9. Sprache & juristische Terminologie (0-3): **3:** Präzise, juristisch sauber, gut verständlich. **2:** Teilweise umständlich/ungenau, aber verständlich. **0-1:** Häufig missverständlich, falsche Begriffe, viele sprachliche Brüche.
10. Formalia: Normzitierweise, Sachverhaltsarbeit, Sorgfalt (0-2): **2:** Normen überwiegend korrekt zitiert; Sachverhalt korrekt verarbeitet; keine groben Schnitzer. **1:** Einzelne Zitier-/Sorgfaltsfehler, nicht gravierend. **0:** Häufige grobe Fehler (Rollen vertauscht, zentrale Fakten falsch wiedergegeben, Normchaos).

Umrechnung Rohpunkte → Notenpunkte (0-18; <4 = nicht bestanden): 0-12→0, 13-25→1, 26-38→2, 39-49→3, 50-53→4, 54-56→5, 57-59→6, 60-63→7, 64-66→8, 67-69→9, 70-73→10, 74-76→11, 77-79→12, 80-83→13, 84-86→14, 87-89→15, 90-93→16, 94-96→17, 97-100→18

Ausgabeformat:

- Gib eine JSON-Ausgabe ohne zusätzlichen Text aus.
- Für jede Dimension: score (0 bis max, ggf. 0.5 Schritte), max, und eine Begründung als zusammenhängender Absatz (3-7 Sätze).
- Danach: total_score (0-100), grade_points (0-18), passed (true/false), kurze Gesamtwürdigung (2-5 Sätze) und 3 konkrete Verbesserungshinweise.

ANGABE / SACHVERHALT: {{{ANGABE_TEXT}}}

MUSTERLÖSUNG: {{{MUSTERLOESUNG_TEXT}}}

STUDIERENDENLÖSUNG: {{{LOESUNG_TEXT}}}

Aufgabe: Bewerte die Lösung nach dem vorgegebenen 100-Punkte-Schema. Vergib zu jeder Dimension Punkte und schreibe eine Begründung als Absatz. Berechne die Summe (0-100) und wandle in Notenpunkte (0-18) um. Markiere passed=true nur, wenn grade_points >= 4.

Gib ausschließlich JSON aus.

English translation (reader reference; not sent to any model):

You are a strict but fair grader (“LLM Judge”) for legal exam-style cases written by German students up to the first state examination (*Erstes Staatsexamen*).

You evaluate exclusively on the basis of: 1. the task/fact pattern, 2. the student solution (*Studierendenlösung*), 3. the reference solution (*Musterlösung*).

Do not use any external sources and do not invent any facts.

Principles:

- The reference solution is the benchmark. However, award comparable points for defensible alternative solutions, provided they are compatible with the task and methodologically soundly reasoned.
- Follow-on-error principle (*Folgefehlerprinzip*): do not punish one error twice. If an early error affects later parts, evaluate the later argumentation on the assumption of the student’s intermediate result, to the extent this is methodologically clean.
- No points for mere buzzwords without definition / subsumption.
- In your justification, refer concretely to what the student solution says and how it fits the reference solution.

Grading scheme (100 points in total):

1. Correctness of result (0-20): **Full points (≈18-20):** core results match the reference solution or are defensibly equivalent (e.g. a different but methodologically correctly reasoned result). **Middle (≈10-14):** the main tendency is right, but one essential result is wrong or the legal consequences are noticeably off. **Low (≈0-8):** central results are missed / inconsistent; the solution misses the point of the case.
2. Completeness and problem identification (0-10): **Full points (≈9-10):** all essential problems from the task / reference solution are recognised; no major “blind spots”. **Middle (≈5-7):** one essential problem is missing or a focal point is overlooked; otherwise sound. **Low (≈0-4):** several central problems are missing, or many irrelevant side topics are examined (“topic essay” instead of case solution).
3. Legal bases and examination systematics (0-10): **Full points (≈9-10):** correct causes of action / examination programmes, logical sequence, clean examination sections. **Middle (≈5-7):** small systematic errors (sequence, individual incorrect / unnecessary bases), but the overall structure holds. **Low (≈0-4):** wrong basic structure (e.g. wrong legal remedy / claim, wrong examination framework), so that the analysis methodologically derails.
4. Legal knowledge (definitions, norms, scholarly debates) (0-15): **Full points (≈13-15):** definitions are correct, norms are apt, scholarly debates appear only where needed; the decision in the debate is reasoned. **Middle (≈8-11):** individual definitions / norms are imprecise; scholarly debates are schematic or patchy, but not wrong in a case-decisive way. **Low (≈0-7):** frequently wrong definitions / norms or invented requirements; scholarly debates are muddled / faulty.
5. Subsumption and quality of argumentation (connection to the case) (0-15): **Full points (≈13-15):** con-

sistent reference to facts, clean subsumption, comprehensible arguments, structured balancing. **Middle (≈8-11):** in part only abstract / “empty-formula-like”, but essential points are still solved with reference to the case. **Low (≈0-7):** hardly any subsumption, predominantly definitions without application; results are unreasoned or contradictory.

6. Setting priorities and depth of analysis (0-10): **Full points (≈9-10):** long fact-pattern blocks / indications are appropriately treated in depth; easy points are kept short; the focus matches the reference solution and the “signals” of the task. **Middle (≈5-7):** either overlength on minor matters or under-weighting of a focal point, but still recognizable exam awareness. **Low (≈0-4):** massive misallocation: the main problem is barely treated, secondary matters dominate.
7. Methodological style: major premise → definition → subsumption → result (0-10): **Full points (≈9-10):** clearly recognizable at all important examination points; intermediate results are drawn; also observed in spirit when written in judgement style. **Middle (≈5-7):** the basic scheme is present but is frequently blended or omitted; definitions or results are occasionally missing. **Low (≈0-4):** no examination-oriented style; only narration, bullet points, or bare assertions of results.
8. Structure, structural levels, and reader guidance (0-5): **5:** stringent structure, clean levels, examination points are findable. **3-4:** small inconsistencies (level jumps, headings not precisely fitting). **0-2:** unstructured block of text; structure is misleading.
9. Language and legal terminology (0-3): **3:** precise, legally clean, easily understandable. **2:** partly clumsy / imprecise, but understandable. **0-1:** frequently misleading, wrong terms, many linguistic breaks.
10. Formalia: norm citation style, work with the fact pattern, care (0-2): **2:** norms cited predominantly correctly; fact pattern processed correctly; no gross blunders. **1:** isolated citation / care errors, not serious. **0:** frequent gross errors (roles swapped, central facts misreported, norm chaos).

Conversion of raw points → grade points (0-18; <4 = fail): 0-12→0, 13-25→1, 26-38→2, 39-49→3, 50-53→4, 54-56→5, 57-59→6, 60-63→7, 64-66→8, 67-69→9, 70-73→10, 74-76→11, 77-79→12, 80-83→13, 84-86→14, 87-89→15, 90-93→16, 94-96→17, 97-100→18

Output format:

- Output a JSON response without any additional text.
- For each dimension: score (0 to max, in 0.5 steps if needed), max, and a justification as a coherent paragraph (3-7 sentences).
- Then: total_score (0-100), grade_points (0-18), passed (true/false), a short overall assessment (2-5 sentences), and 3 concrete suggestions for improvement.

TASK / FACT PATTERN: {{{ANGABE_TEXT}}}

REFERENCE SOLUTION: {{{MUSTERLOESUNG_TEXT}}}

STUDENT SOLUTION: {{{LOESUNG_TEXT}}}

Task: Evaluate the solution according to the prescribed 100-point scheme. Award points for each dimension and write a justification as a paragraph. Calculate the sum (0-100) and convert it into grade points (0-18). Set passed=true only if grade_points >= 4.

Output only JSON.

L.2 Doctrinal Principles (Grundprinzipien)

Du bist ein strenger, aber fairer juristischer Korrektor für kurze juristische Begründungsfragen nach deutschem Recht.

Du bewertest ausschließlich auf Basis von:

1. Fall
2. Fragestellung
3. Referenzlösung
4. Zu bewertende Antwort

Nutze keine externen Quellen und erfinde keine Tatsachen.

Bewertungsgrundsätze

- Die Referenzlösung ist Orientierung, nicht die einzig mögliche vertretbare Lösung.
- Vergib volle oder nahezu volle Punkte auch für methodisch vertretbare Alternativbegründungen.
- Entscheidend ist die juristische Richtigkeit und Qualität der Begründung, nicht die sprachliche Ähnlichkeit zur Referenz.
- Keine Punkte für bloße Schlagworte ohne nachvollziehbare Begründung.
- Eine falsche Ja/Nein-Entscheidung begrenzt die Maximalpunktzahl erheblich, selbst wenn Teile der Begründung juristisch plausibel sind.

Bewertungsschema (100 Punkte)

1. Ergebnisrichtigkeit (0-40)

- Ist die Ja/Nein-Entscheidung juristisch zutreffend oder vertretbar?

1. Rechtskenntnis und Normbezug (0-25)

- Werden einschlägige Normen, Definitionen oder Prinzipien korrekt erkannt und verwendet?

1. Subsumtion und Fallbezug (0-25)

- Wird der konkrete Sachverhalt nachvollziehbar unter die rechtlichen Voraussetzungen subsumiert?

1. Klarheit und Präzision (0-10)

- Ist die Antwort präzise, verständlich und methodisch sauber formuliert?

Ausgabeformat Gib ausschließlich gültiges JSON aus:

```
{
  ”scores”: {
    ”result_correctness”: {
      ”score”: ...,
      ”max”: 40,
      ”reason”: ”...”
    },
  ”legal_knowledge”: {
    ”score”: ...,
    ”max”: 25,
    ”reason”: ”...”
  },
”subsumption”: {
  ”score”: ...,
  ”max”: 25,
  ”reason”: ”...”
},
```

```
”clarity”: {
  ”score”: ...,
  ”max”: 10,
  ”reason”: ”...”
}
},
”total_score”: ...,
”overall_assessment”: ”...”
}
```

Fall {{fall}}

Fragestellung {{task}}

Referenzlösung Antwort: {{binary_solution}}
Begründung: {{reasoning}}

Zu bewertende Antwort {{answer}}

English translation (reader reference; not sent to any model):

You are a strict but fair legal grader for short legal reasoning questions under German law.

You evaluate exclusively on the basis of:

1. the case,
2. the question,
3. the reference solution,
4. the answer to be evaluated.

Do not use any external sources and do not invent any facts.

Grading principles

- The reference solution is a guide, not the only defensible solution.
- Award full or near-full points also for methodologically defensible alternative justifications.
- What matters is the legal correctness and quality of the reasoning, not linguistic similarity to the reference.
- No points for mere buzzwords without a comprehensible justification.
- A wrong Ja/Nein decision substantially limits the maximum score, even if parts of the reasoning are legally plausible.

Grading scheme (100 points)

1. Correctness of result (0-40)

- Is the Ja/Nein decision legally correct or defensible?

1. Legal knowledge and reference to norms (0-25)

- Are the applicable norms, definitions, or principles correctly identified and used?

1. Subsumption and reference to the case (0-25)

- Is the concrete fact pattern subsumed under the legal requirements in a comprehensible way?

1. Clarity and precision (0-10)

- Is the answer precisely, understandably, and methodologically cleanly formulated?

Output format Output only valid JSON:

```
{
  ”scores”: {
    ”result_correctness”: {
      ”score”: ...,
      ”max”: 40,
      ”reason”: ”...”
    },
  ”legal_knowledge”: {
    ”score”: ...,
    ”max”: 25,
    ”reason”: ”...”
  },
”subsumption”: {
  ”score”: ...,
  ”max”: 25,
  ”reason”: ”...”
},
”clarity”: {
  ”score”: ...,
  ”max”: 10,
  ”reason”: ”...”
}
},
”total_score”: ...,
”overall_assessment”: ”...”
}
```

Case {{fall}}

Question {{task}}

Reference solution Answer: {{binary_solution}}
Reasoning: {{reasoning}}

Answer to be evaluated {{answer}}

M LLM-judge configuration

The primary judge GPT-5.4-mini is the one used for every reported leaderboard score and for the RQ5 within-judge stochasticity analysis (k=3 repeats per generation). The five secondary judges Opus-4.7, Gemini-3.1-Pro, DeepSeek-V4-Pro, Qwen3.5-397B-A17B, and Sonnet-4.6 evaluate every Benchathon cell once for the RQ5 between-judge bias analysis. The full model ids are shown in the *Model* column below; see Table 10 (Appendix B) for the canonical alias-to-id mapping.

The full judge prompts are reproduced in Appendix L: the exam-case rubric and the Doctrinal Principles rubric. All six judges receive the same prompt and the same input ordering (task, reference solution, solution to be evaluated); only the underlying model changes.

N Judge calibration - supplementary detail

This appendix collects the supplementary numerical detail behind the RQ5 per-judge calibration analysis: the judge-swapped system leaderboard and its rank agreement in Tables 19 and 20, the six-judge calibration table referenced from Figure 6, the within-judge per-pass breakdown of the primary GPT-5.4-mini judge in Table 22, the per-annotator alt-test p-values, and a sanity check on the intersection of picks where the Config-B cross-validation judges scored.

Judge-swapped leaderboard. On the cross-judge Benchathon subset (every leaderboard system covered; per-system n in Table 20), each of the five cross-validation judges re-scores the same generations as the primary judge. Table 19 reports the system-level rank agreement of each judge's per-system means against the primary judge's on the identical subset, together with each judge's mean pass rate at the fixed 50-point threshold, raw and after removing that judge's measured offset against the blind human pool (Table 21). Strictness shifts pass rates at the fixed threshold; the offset correction largely restores them, and the ordering itself is stable under every judge.

Per-judge calibration. Table 21 reports the per-judge calibration statistics behind Figure 6. It gives two quantities. The direct judge-vs-pool offset $\bar{\Delta}_{\text{dir}} = \overline{\text{judge} - \text{full}}$ is the number to quote when prose says "the judge sits X raw points above the pool' '. The Calderon pool-substitution shift $\bar{\Delta}_{\text{sub}} = \overline{\text{sub} - \text{full}}$ is attenuated by $1/k$ at $k = 3$ raters. Both come with a paired t-test. Shapiro–Wilk does not reject normality in all but two cells; in those two the near-zero offset is unchanged under a Wilcoxon signed-rank cross-check, so the parametric test applies throughout.

Per-pass detail. Table 22 reports the same per-judge calibration statistics as Table 21, but with the three Config-A GPT-5.4-mini intra-judge passes broken out individually alongside the baseline single-pass. This gives a $k = 4$ within-judge stochasticity check on the primary judge. The four per-pass offsets cluster with a stdev of 1.5 raw points on human-content picks and 1.7 on LLM-content picks. That is small relative to the cross-judge leniency gradient (the lenient judges sit +6 to +18 raw points above the pool on LLM picks) and to the ≈,24-raw-point within-solution max-min range across the blind reviewers. But it is not strictly deterministic.

Per-annotator alt-test detail. Tables 23 and 24 report the full Calderon blind-pool alt-test by judge for the human-authored and LLM-generated picks respectively. Each block's header states the winning rate ω that block accumulates. The rows below show the five blind reviewers' per-annotator n_j, ρ_j^f, ρ_j^h, $\bar{d}_j$, one-sided p-value (Wilcoxon signed-rank throughout, since every $n_j < 30$), and whether the Benjamini-Yekutieli FDR procedure at $q = 0.05$ across $m = 5$ rejected the null. Only annotators flagged ★ contribute to ω.

Intersection sanity check. Opus-4.7's Benchathon coverage is $n = 28$ on the human-pick subset rather than $n = 30$, because Opus-4.7 did not produce a parseable score on two picks (likely safety filter or output-shape failure). Restricting the alt-test to the intersection of picks every judge scored ($n = 28$) preserves the per-judge ordering and the two-judge headline: GPT-5.4-mini and Opus-4.7 still clear $\omega \geq 0.5$. This makes Sonnet-4.6 and Gemini-3.1-Pro's "near pass" at $\omega = 0.40$ on the full $n = 30$ marginally sample-dependent on the two picks Opus-4.7 missed. We report the full-n ω in the main table and treat the intersection as a robustness anchor only.

N.1 Creator-anchored bias signature and corrected-alt-test sensitivity

The main-text alt-test ranks the judges by their per-solution offset against the blind-reviewer pool. GPT-5.4-mini and Opus-4.7 sit close to the pool and clear the substitution bar, while the lenient judges fail. This subsection asks a complementary question. Does an *independent* expert anchor reproduce the same offset ranking, or is the pool-anchored signature an artefact of the specific blind reviewers we drew? We use the un-blind creator grade as the independent anchor. Creators (i) hold privileged knowledge of the intended solution, (ii) are the most legally expert graders on the roster, and (iii) are excluded from the IRR pool by design, so they do not contaminate the blind-pool alt-test.

Method. For every (judge, pick) we estimate a leave-one-out per-judge correction against the creator grade in two tiers reported side-by-side: a single-parameter *scalar* shift $y' = y - \overline{(y_{\text{judge}} - y_{\text{creator}})}$ and a two-parameter *affine* fit $y' = \hat{a} + \hat{b}\, y_{\text{judge}}$ via OLS of creator on judge. We then re-run the blind-pool alt-test pipeline byte-identically with calibrated scores substituted, so the only change relative to the headline Tables 23 and 24 is the per-pick judge value. Two leak-clean variants are reported: **grader-leave-one-out**

(GLOO), in which a tested annotator's creator picks are additionally excluded from the calibration fit (graders 01 and 04 wear both creator and blind-reviewer hats across the roster, and a naive LOO would style-train the judge on the very annotator the alt-test then asks it to replace), and **pure-creator**, restricting the anchor to picks whose creator is grader_03 or grader_06 – the two graders who never appear in the blind pool – which is structurally leak-free at the cost of a smaller anchor ($n \approx 30$ vs. $n \approx 45$). Calibration is fit pooled across human + LLM picks. A pool-specific refit is reported as sensitivity. A stacked bootstrap (resample anchor *and* test set, refit, re-run alt-test, 500 reps) propagates both calibration and test-set variance into a 95% percentile CI on ω.

Bias-signature finding. Per-judge LOO scalar coefficients against the creator under GLOO are tightly clustered for GPT-5.4-mini ($\hat{a}_{\text{scalar}} \approx -0.3$) and Opus-4.7 ($\hat{a}_{\text{scalar}} \approx +0.5$) – both essentially perfectly calibrated to the strongest grader on the roster – while Sonnet-4.6 sits at ≈ -3.5, DeepSeek-V4-Pro at ≈ -4, Gemini-3.1-Pro at ≈ -11, and Qwen3.5-397B-A17B at ≈ -12 (the negative sign means the judge is lenient relative to the creator – equivalently the judge mean sits 3.5 to 12 raw points above the creator mean). The affine slope $\hat{b}$ is sub-unity for every judge ($\hat{b} \in [0.77, 0.93]$), so the judges also use a slightly compressed range relative to the creator. The pool-anchored ranking from Table 21 and the creator-anchored ranking agree on the strictness order from two structurally independent reference signals – the strongest evidence that the gradient is a property of the judges, not of either grading subgroup. Per-annotator $(\hat{a}, \hat{b})$ spreads across the five blind reviewers under GLOO are small ($\hat{a}_{\text{scalar}}$ spread ≤ 1.4 raw points, $\hat{b}_{\text{affine}}$ spread ≤ 0.14), confirming the GLOO fit is internally stable.

Corrected-ω finding. Under leak-clean GLOO calibration, the point ω lifts for some judges (e.g. Qwen3.5-397B-A17B scalar from 0.00 to 0.60 on human picks; Sonnet-4.6 scalar to 0.80) but not uniformly – affine correction destabilises Opus-4.7's point estimate (from 0.60 raw to 0.00 affine), reflecting that a near-identity transform on an already-calibrated judge can still flip per-instance verdicts at the $m = 5$ resolution. More importantly, **no stacked-bootstrap lower bound clears $\omega = 0.5$ for any judge × pool × tier – including the two judges that already pass at raw scores.** At $m = 5$ blind reviewers and $n = 30$ picks the alt-test's discrete ω quantum is 0.20, so its bootstrap distribution under resampling spans most of $[0, 1]$ for any cell. The strongest signal in the matrix is GPT-5.4-mini's raw-score stacked-bootstrap lower bound of 0.20, which is what motivates the main-text claim. Corrected ω does not improve on it. We therefore treat creator-anchored calibration as quantifying the per-judge bias *signature* and explaining the inter-judge alt-test gradient, not as a calibration-recovery argument.

Pure-creator vs. GLOO sensitivity. The two leak-clean anchor strategies disagree on point ω: pure-creator yields more PASSes at the point estimate (e.g. GPT-5.4-mini scalar $\omega = 1.00$, Gemini-3.1-Pro scalar $\omega = 1.00$, DeepSeek-V4-Pro affine $\omega = 1.00$) because graders 03 and 06's grading style sits closer to the average blind reviewer than graders 01 and 04's – i.e., the *choice* of expert anchor materially affects the corrected verdict. Treat the GLOO result as the methodologically conservative reading, since it uses a strictly larger and more diverse anchor and only excludes information where strictly necessary for cleanness.

LLM-pick robustness. Across every calibration tier, anchor strategy, and $\varepsilon \in \{0.15, 0.20\}$ examined here, LLM-pick ω stays at $\{0.00, 0.20\}$ for every judge. The LLM-pick failure is therefore residual disagreement about what makes a good LLM-generated solution, not a calibration artefact – a finding that constrains the Limitations discussion of judge-human divergence on AI-styled writing rather than rescuing it.

O Error-analysis mode counts

Per-mode instance counts and per-tier rates behind the Error analysis section, computed at corpus scale from the primary-judge scores by the released pipeline (scripts/derive_error_analysis.py). The two *Falllösung* corpora share the five flagged modes; Doctrinal Principles uses the three decision-centric modes of its 4-dimension rubric.

P Reasoning-mode ablation

A controlled test of the reasoning mechanism on open weights: Qwen3-235B-A22B-*Instruct*-2507, the official non-thinking counterpart of the Thinking-2507 variant in the main evaluation, generated the full Benchathon subset (both prompt structures) and the full Doctrinal Principles corpus through the identical pipeline – same managed inference provider, same German prompts, same decoding settings (temperature 0.6; max output tokens 16,000 on Benchathon, 8,000 on Doctrinal Principles), and the same primary GPT-5.4-mini judge configurations. Table 27 pairs the two variants. The ablation system is not part of the 12-system leaderboard; its rows live in a released sidecar file (data/raw/ablation/).

The toggle's effect is modest relative to tier gaps. Reasoning-on scores +3.5 raw points on Benchathon and +4.1 on Doctrinal Principles – against a gap of roughly 28 raw points from this system to the Benchathon flagship cluster. On the short-form corpus the gain costs about 5× the latency; on Benchathon the non-thinking variant spends its budget on roughly twice the visible output instead. An explicit reasoning mechanism alone therefore does not close the tier gap on subsumption-based legal reasoning – consistent with the RQ1 observation that the reasoning-enabled Qwen3-235B underperforms its size.

Q Glossary of German terms

German terms of art are kept alongside English glosses because they lack exact English equivalents and appear verbatim in the released German-language prompts and rubric. Table 28 collects every German term used in the paper.

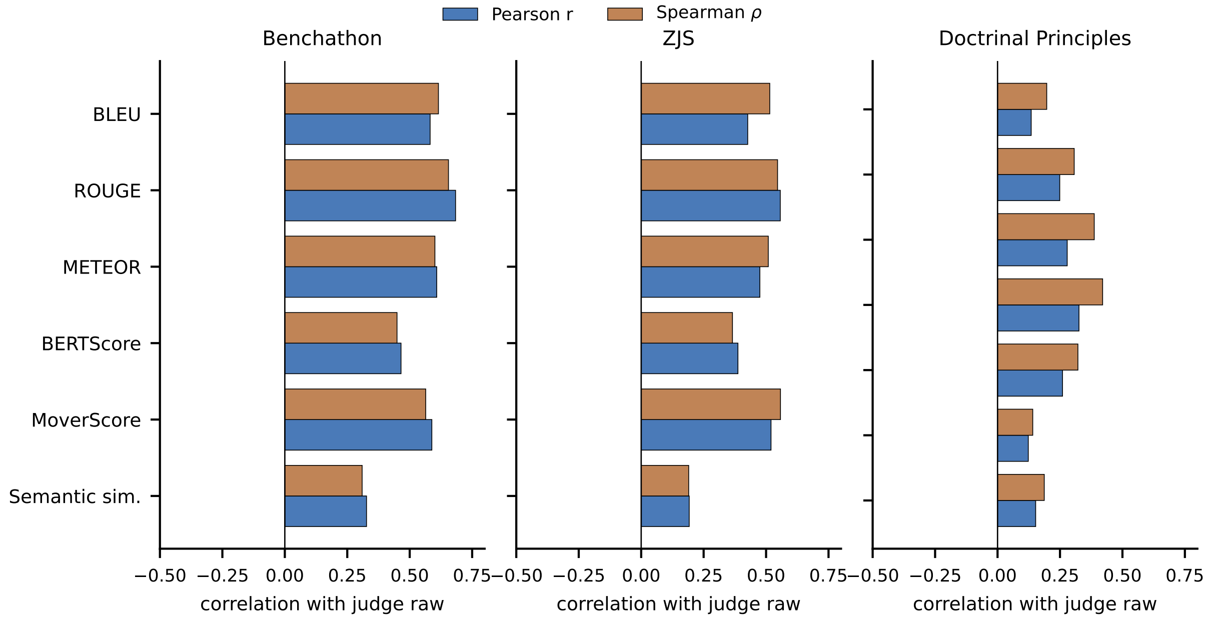

Figure 4: Per-generation correlation between each automatic metric and the LLM-judge raw score on the Benchathon (left), ZJS (middle), and Doctrinal Principles (right) corpora. Bars report Pearson r and Spearman ρ, with n varying per metric. Metrics are shown in the fixed canonical order shared with the per-corpus correlation table in this appendix. Judge on Doctrinal Principles is the custom 4-dimension rubric.

Table 17: Agreement statistics on the Benchathon human-grading validation subset (design: three blind reviewers per solution with one author-informed creator review reported separately). Judge-vs-human rows report Pearson r, Spearman ρ, MAE, and pass/fail Cohen's κ, split by solution provenance (human-authored vs. LLM-generated). Human IRR rows report ICC(2,1) raw, ICC(2,k) raw, and ICC(2,k) grade-points. The pooled row covers all 45 picks, the indented rows are per-solution-type sub-pools (15 picks each). Calderon alt-test rows report the per-annotator winning rate ω and average advantage probability ρ at the stated ε, with Benjamini-Yekutieli FDR at $q = 0.05$ across the m blind reviewers. $\omega \geq 0.5$ is the alt-test pass criterion. Blind-pool rows use the three blind reviewers as reference. Creator-reference rows use the un-blind creator grade as a single-expert reference.

Comparison	Sample	Stat 1	Stat 2	Stat 3
Judge vs. mean-human (human sol.)	$n = 30$	$r = 0.76$, $\rho = 0.76$	MAE raw 10.8, MAE gp 2.69	$\kappa = 0.60$
Judge vs. mean-human (LLM sol.)	$n = 15$	$r = 0.64$, $\rho = 0.49$	MAE raw 13.1, MAE gp 2.87	$\kappa = 0.40$
Human IRR pooled (k=3 blind, all 45)	$n = 45 \times 3$	ICC(2,1) raw 0.63	ICC(2,k) raw 0.84	ICC(2,k) gp 0.79
traditional human	$n = 15 \times 3$	ICC(2,1) raw 0.71	ICC(2,k) raw 0.88	ICC(2,k) gp 0.83
co-creation human	$n = 15 \times 3$	ICC(2,1) raw 0.46	ICC(2,k) raw 0.71	ICC(2,k) gp 0.65
LLM-generated	$n = 15 \times 3$	ICC(2,1) raw 0.69	ICC(2,k) raw 0.87	ICC(2,k) gp 0.83
Calderon blind-pool alt-test (human sol.)	$m = 5 \times n = 30$	$\omega = 0.60$ ($\varepsilon = 0.15$)	$\rho = 0.62$	passes
Calderon blind-pool alt-test (LLM sol.)	$m = 5 \times n = 15$	$\omega = 0.00$ ($\varepsilon = 0.15$)	$\rho = 0.41$	fails
Calderon creator-reference alt-test (human sol., $\varepsilon = 0.20$)	$m = 5 \times n = 30$	$\omega = 0.60$ ($\varepsilon = 0.20$)	$\rho = 0.55$	passes
Calderon creator-reference alt-test (LLM sol., $\varepsilon = 0.20$)	$m = 5 \times n = ---$	$\omega = 0.00$ ($\varepsilon = 0.20$)	$\rho = 0.47$	fails

Table 18: LLM-judge configuration. Temperature and max output tokens are the values captured in the per-call metadata of the Benchathon evaluation runs (primary 2026-05-31, cross-validation 2026-05-21 and -31); the retry budget is the worker-side default of 3, with the observed retry count across each run reported in parentheses. No provider-side snapshot id is returned for these aliases; the model id plus run date pins reproducibility.

Role	Model	Run date	T	Max tok.	Retry budget
Primary judge	gpt-5.4-mini	2026-05-31	1.0	8000	3 (obs. 0)
Cross-validation (Anthropic)	claude-opus-4-7	2026-05-21	1.0	8000	3 (obs. 0)
Cross-validation (Anthropic)	claude-sonnet-4-6	2026-05-31	1.0	8000	3 (obs. 0)
Cross-validation (Google)	gemini-3.1-pro-preview	2026-05-21	1.0	8000	3 (obs. 0)
Cross-validation (DeepSeek)	deepseek-ai/DeepSeek-V4-Pro	2026-05-31	1.0	8000	3 (obs. 0)
Cross-validation (Alibaba)	Qwen/Qwen3.5-397B-A17B	2026-05-31	1.0	8000	3 (obs. 0)

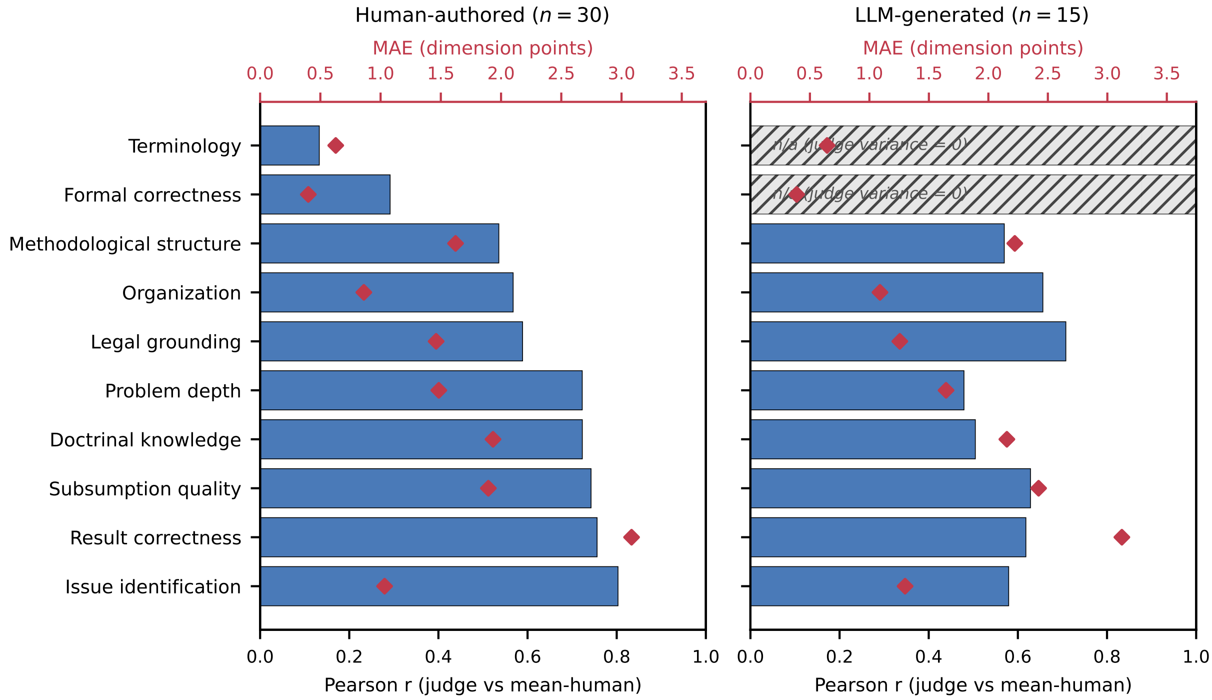

Figure 5: Per-rubric-dimension agreement between the LLM judge and the mean of the blind human reviewers, split by solution provenance. Left: human-authored solutions (n = 30). Right: LLM-generated solutions (n = 15). Bars: Pearson r (bottom axis). Points: MAE in dimension points (top axis). Dimension ordering is shared across panels (sorted by human-side Pearson). Hatched bars marked n/a indicate that the judge returned a constant score across the 15 LLM picks on that dimension, leaving the Pearson statistic mathematically undefined (judge variance = 0). The MAE is still well-defined and is plotted.

Table 19: System-level rank agreement of each cross-validation judge against the primary judge on the shared Benchathon subset (Spearman ρ / Kendall τ over 12 per-system means), and each judge's mean pass rate at the fixed $\geq$ 50/100 threshold, raw and after removing the judge's measured offset vs. the blind human pool.

Judge	ρ	τ	Pass (raw)	Pass (offset-corr.)
GPT-5.4-mini (primary)	—	—	61%	61%
Opus-4.7	0.97	0.88	58%	55%
Sonnet-4.6	0.95	0.85	70%	58%
Gemini-3.1-Pro	0.94	0.82	73%	45%
DeepSeek-V4-Pro	0.94	0.84	60%	52%
Qwen3.5-397B	0.94	0.82	87%	58%

Table 20: Judge-swapped Benchathon leaderboard: per-system mean raw score under each judge on the identical generation subset, systems ordered by the primary judge. Strictness moves the level; the ordering is preserved (rank agreement in Table 19).

System	n	GPT-5.4-mini (primary)	Opus-4.7	Sonnet-4.6	Gemini-3.1-Pro	DeepSeek-V4-Pro	Qwen3.5-397B
Opus-4.7	15	69.3	71.7	74.7	82.8	76.5	81.5
Gemini-3.1-Pro	15	68.3	68.4	69.6	81.7	74.6	76.3
GPT-5.4	30	68.2	69.6	71.9	80.9	72.0	80.9
Gemini-3.1-Flash-Lite	15	63.5	63.1	67.1	82.0	68.1	77.3
Sonnet-4.6	15	58.9	66.0	70.3	75.9	74.6	77.4
GPT-5.4-mini	15	58.6	51.1	57.3	60.3	50.0	65.5
DeepSeek-V4-Pro	15	56.1	56.9	62.5	70.1	60.5	71.4
Qwen3.5-122B	15	49.4	42.8	48.5	54.6	49.6	59.6
DeepSeek-V4-Flash	15	49.3	47.8	55.3	62.2	52.9	65.1
Qwen3.6-35B	15	42.4	31.5	44.2	40.2	40.7	54.2
Qwen3-235B	28	41.3	30.7	41.2	43.1	36.1	51.9
Llama-4	14	37.7	29.0	37.0	37.8	31.6	51.8

Table 21: Per-judge calibration on the two blind-graded subsets of the Benchathon validation set, ordered by increasing leniency on human-content picks. Direct offset $\bar{\Delta}_{dir}$ is the per-pick paired difference between the judge and the mean of three blind reviewers. $\bar{\Delta}_{sub}$ is the same quantity after the Calderon §1 pool substitution and is attenuated by 1/3 at $k = 3$ raters. SW p tests normality of the paired sub–full differences. Six judges: GPT-5.4-mini is the primary judge that anchors RQ1–RQ2; Opus-4.7, Sonnet-4.6, DeepSeek-V4-Pro, Gemini-3.1-Pro and Qwen3.5-397B-A17B are Benchathon-only cross-validation judges.

Configuration	n	$\bar{\Delta}_{dir}$ (sd)	t (p)	$\bar{\Delta}_{sub}$ (sd)	SW p	t_{sub} (p)
Human picks: Baseline GPT-5.4-mini (primary)	$n = 30$	−0.01 (sd 13.43)	−0.00 (0.9964)	0.20 (sd 4.99)	0.02	0.22 (0.8279)
Opus-4.7	$n = 28$	−2.06 (sd 13.93)	−0.78 (0.4409)	−0.52 (sd 4.71)	0.78	−0.59 (0.5611)
Sonnet-4.6	$n = 30$	0.62 (sd 12.96)	0.26 (0.7945)	0.41 (sd 4.46)	0.94	0.50 (0.6178)
DeepSeek-V4-Pro	$n = 30$	2.11 (sd 14.89)	0.77 (0.4448)	0.91 (sd 4.69)	0.99	1.06 (0.2987)
Gemini-3.1-Pro	$n = 30$	6.82 (sd 15.25)	2.45 (0.0206)	2.48 (sd 4.79)	0.96	2.83 (0.0084)
Qwen3.5-397B-A17B	$n = 30$	8.72 (sd 15.00)	3.19 (0.0034)	3.11 (sd 4.91)	0.31	3.47 (0.0017)
LLM picks: Baseline GPT-5.4-mini (primary)	$n = 15$	−0.47 (sd 16.24)	−0.11 (0.9130)	0.83 (sd 7.65)	0.33	0.42 (0.6795)
Opus-4.7	$n = 15$	1.37 (sd 10.95)	0.48 (0.6362)	1.44 (sd 5.62)	0.52	0.99 (0.3368)
Sonnet-4.6	$n = 15$	6.87 (sd 13.25)	2.01 (0.0645)	3.28 (sd 6.62)	0.85	1.92 (0.0757)
DeepSeek-V4-Pro	$n = 15$	5.83 (sd 16.98)	1.33 (0.2045)	2.93 (sd 6.63)	0.48	1.71 (0.1088)
Gemini-3.1-Pro	$n = 15$	18.17 (sd 8.40)	8.37 (0.0000)	7.04 (sd 5.03)	0.89	5.42 (0.0001)
Qwen3.5-397B-A17B	$n = 15$	16.67 (sd 13.47)	4.79 (0.0003)	6.54 (sd 6.03)	0.32	4.20 (0.0009)

Table 22: Within-judge stochasticity detail for the primary GPT-5.4-mini judge. The Baseline row is the single-pass run that anchors RQ1–RQ2. The Config A block adds three intra-judge re-runs (3×) so the four offsets in each block give a $k = 4$ within-judge stochasticity check. The per-pass offset $\bar{\Delta}_{dir}$ stdev across the four passes is 1.53 raw points on human picks and 1.65 on LLM picks, with max $|\text{pass} - \text{mean}|$ of 2.23 (human) and 2.11 (LLM). Same columns as Table 21.

Configuration	n	$\bar{\Delta}_{dir}$ (sd)	t (p)	$\bar{\Delta}_{sub}$ (sd)	SW p	t_{sub} (p)
Human picks: Baseline GPT-5.4-mini (1×, primary)	$n = 30$	−0.01 (sd 13.43)	−0.00 (0.9964)	0.20 (sd 4.99)	0.02	0.22 (0.8279)
Config A GPT-5.4-mini (3× mean)	$n = 30$	−1.68 (sd 14.43)	−0.64 (0.5278)	−0.36 (sd 5.41)	0.10	−0.36 (0.7202)
pass 1	$n = 30$	−0.88 (sd 18.22)	−0.26 (0.7937)	−0.09 (sd 6.62)	0.02	−0.07 (0.9419)
pass 2	$n = 30$	−0.68 (sd 13.82)	−0.27 (0.7901)	−0.02 (sd 5.31)	0.10	−0.02 (0.9819)
pass 3	$n = 30$	−3.49 (sd 13.57)	−1.41 (0.1692)	−0.96 (sd 5.00)	0.57	−1.05 (0.3010)
LLM picks: Baseline GPT-5.4-mini (1×, primary)	$n = 15$	−0.47 (sd 16.24)	−0.11 (0.9130)	0.83 (sd 7.65)	0.33	0.42 (0.6795)
Config A GPT-5.4-mini (3× mean)	$n = 15$	2.34 (sd 14.64)	0.62 (0.5450)	1.77 (sd 7.05)	0.38	0.97 (0.3475)
pass 1	$n = 15$	2.97 (sd 17.50)	0.66 (0.5222)	1.98 (sd 7.85)	0.45	0.98 (0.3455)
pass 2	$n = 15$	1.13 (sd 14.89)	0.29 (0.7724)	1.37 (sd 6.77)	0.12	0.78 (0.4472)
pass 3	$n = 15$	2.93 (sd 14.04)	0.81 (0.4319)	1.97 (sd 7.15)	0.89	1.07 (0.3046)

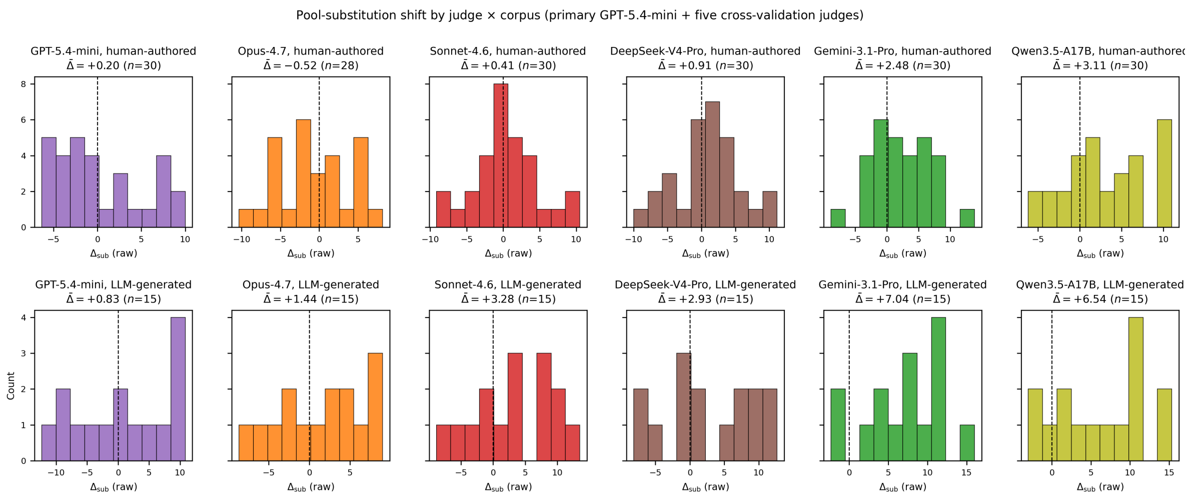

Figure 6: Pool-substitution shift $\bar{\Delta}_{sub} = \overline{sub} - \overline{full}$ per judge × corpus, across the six judges that score the Benchathon validation subset. For each pick, full is the mean of three blind reviewers and sub is the mean of two blind reviewers plus the LLM judge; by construction $\bar{\Delta}_{sub} \approx \bar{\Delta}_{dir}/3$ where $\bar{\Delta}_{dir} = \overline{\text{judge} - \text{full}}$ is the direct judge-vs-pool offset reported in Table 21. Top row: human-authored picks ($n = 30$ for every judge except Opus, where $n = 28$ because two evaluations did not yield a parseable score, likely a safety-filter or output-shape failure). Bottom row: LLM-generated picks ($n = 15$). The primary GPT-5.4-mini and Opus-4.7 sit essentially on the blind-reviewer pool mean ($p > 0.3$ on both, not significantly different); Sonnet-4.6 and DeepSeek-V4-Pro are close on human-content picks and drift mildly lenient on LLM picks; Gemini-3.1-Pro and Qwen3.5-397B-A17B are systematically lenient ($p < 0.01$ on both). Most paired $\bar{\Delta}_{sub}$ distributions pass Shapiro–Wilk normality ($p > 0.05$; the two exceptions are unchanged under a Wilcoxon cross-check), so the reported t-test is the appropriate inferential test (see Table 21).

Table 23: Human-authored picks: Calderon §3 alt-test detail by judge. Each block lists the five blind reviewers in the roster (graders 01, 02, 04, 05, 07) for one judge, with that block's headline ω stated in its italic header. ρ_j^f = mean over $i \in I_j$ of $\mathbf{1}\{S(f, x_i, j) \geq S(h_j, x_i, j)\}$ (probability the LLM judge aligns with the remaining $H_i \setminus \{j\}$ at least as closely as h_j); ρ_j^h symmetrically. $\bar{d}_j = \rho_j^h - \rho_j^f$ is the per-annotator advantage probability gap. p-value is one-sided Wilcoxon signed-rank against $\bar{d}_j \geq \varepsilon = 0.15$. ★ marks rejection under Benjamini-Yekutieli FDR at $q = 0.05$ across $m = 5$.

	n_j	ρ^f/ρ^h	$\bar{d}$	p	rej.
Baseline GPT-5.4-mini (primary) ($\omega = 0.60$ passes)					
grader_01	14	0.57 / 0.43	−0.143	0.0247	
grader_02	19	0.68 / 0.32	−0.368	0.0008	★
grader_04	15	0.73 / 0.33	−0.400	0.0034	★
grader_05	21	0.62 / 0.38	−0.238	0.0021	★
grader_07	21	0.48 / 0.52	0.048	0.0444	
Opus-4.7 ($\omega = 0.60$ passes)					
grader_01	13	0.54 / 0.46	−0.077	0.0471	
grader_02	17	0.59 / 0.41	−0.176	0.0101	★
grader_04	15	0.60 / 0.40	−0.200	0.0128	★
grader_05	19	0.58 / 0.42	−0.158	0.0080	★
grader_07	20	0.40 / 0.60	0.200	0.1650	
Sonnet-4.6 ($\omega = 0.40$)					
grader_01	14	0.50 / 0.50	0.000	0.0676	
grader_02	19	0.63 / 0.42	−0.211	0.0070	★
grader_04	15	0.53 / 0.47	−0.067	0.0365	
grader_05	21	0.67 / 0.33	−0.333	0.0007	★
grader_07	21	0.52 / 0.48	−0.048	0.0175	
DeepSeek-V4-Pro ($\omega = 0.20$)					
grader_01	14	0.64 / 0.50	−0.143	0.0453	
grader_02	19	0.68 / 0.32	−0.368	0.0008	★
grader_04	15	0.53 / 0.60	0.067	0.1651	
grader_05	21	0.52 / 0.52	0.000	0.0411	
grader_07	21	0.43 / 0.57	0.143	0.1015	
Gemini-3.1-Pro ($\omega = 0.40$)					
grader_01	14	0.71 / 0.36	−0.357	0.0067	★
grader_02	19	0.74 / 0.26	−0.474	0.0003	★
grader_04	15	0.53 / 0.53	0.000	0.0844	
grader_05	21	0.52 / 0.52	0.000	0.0411	
grader_07	21	0.38 / 0.62	0.238	0.2060	
Qwen3.5-397B-A17B ($\omega = 0.00$)					
grader_01	14	0.50 / 0.57	0.071	0.1479	
grader_02	19	0.47 / 0.53	0.053	0.0567	
grader_04	15	0.40 / 0.60	0.200	0.2106	
grader_05	21	0.33 / 0.67	0.333	0.3667	
grader_07	21	0.33 / 0.67	0.333	0.3667	

Table 24: LLM-generated picks: Calderon §3 alt-test detail by judge. Each block lists the five blind reviewers in the roster (graders 01, 02, 04, 05, 07) for one judge, with that block's headline ω stated in its italic header. ρ_j^f = mean over $i \in I_j$ of $\mathbf{1}\{S(f, x_i, j) \geq S(h_j, x_i, j)\}$ (probability the LLM judge aligns with the remaining $H_i \setminus \{j\}$ at least as closely as h_j); ρ_j^h symmetrically. $\bar{d}_j = \rho_j^h - \rho_j^f$ is the per-annotator advantage probability gap. p-value is one-sided Wilcoxon signed-rank against $\bar{d}_j \geq \varepsilon = 0.15$. ★ marks rejection under Benjamini-Yekutieli FDR at $q = 0.05$ across $m = 5$.

	n_j	ρ^f / ρ^h	$\bar{d}$	p	rej.
Baseline GPT-5.4-mini (primary) ($\omega = 0.00$)					
grader_01	7	0.29 / 0.71	0.429	0.5938	
grader_02	11	0.09 / 0.91	0.818	0.9790	
grader_04	9	0.67 / 0.33	−0.333	0.0273	
grader_05	9	0.56 / 0.44	−0.111	0.0820	
grader_07	9	0.44 / 0.56	0.111	0.2129	
Opus-4.7 ($\omega = 0.20$)					
grader_01	7	0.29 / 0.71	0.429	0.5938	
grader_02	11	0.55 / 0.55	0.000	0.1392	
grader_04	9	0.89 / 0.11	−0.778	0.0039	★
grader_05	9	0.78 / 0.22	−0.556	0.0098	
grader_07	9	0.67 / 0.44	−0.222	0.0645	
Sonnet-4.6 ($\omega = 0.00$)					
grader_01	7	0.00 / 1.00	1.000	1.0000	
grader_02	11	0.36 / 0.73	0.364	0.5845	
grader_04	9	0.67 / 0.33	−0.333	0.0273	
grader_05	9	0.67 / 0.33	−0.333	0.0273	
grader_07	9	0.56 / 0.44	−0.111	0.0820	
DeepSeek-V4-Pro ($\omega = 0.00$)					
grader_01	7	0.00 / 1.00	1.000	1.0000	
grader_02	11	0.36 / 0.64	0.273	0.3501	
grader_04	9	0.78 / 0.22	−0.556	0.0098	
grader_05	9	0.78 / 0.44	−0.333	0.0371	
grader_07	9	0.33 / 0.67	0.333	0.4551	
Gemini-3.1-Pro ($\omega = 0.00$)					
grader_01	7	0.14 / 0.86	0.714	0.8906	
grader_02	11	0.00 / 1.00	1.000	1.0000	
grader_04	9	0.56 / 0.44	−0.111	0.0820	
grader_05	9	0.22 / 0.78	0.556	0.7520	
grader_07	9	0.44 / 0.56	0.111	0.2129	
Qwen3.5-397B-A17B ($\omega = 0.00$)					
grader_01	7	0.00 / 1.00	1.000	1.0000	
grader_02	11	0.18 / 0.91	0.727	0.9731	
grader_04	9	0.56 / 0.44	−0.111	0.0820	
grader_05	9	0.33 / 0.67	0.333	0.4551	
grader_07	9	0.33 / 0.67	0.333	0.4551	

Table 25: Error-analysis mode counts on the two *Falllösung* corpora under the primary judge: flagged instances per mode, corpus n, and per-tier flag rates.

Corpus	Mode	Count	n	Flagship	Efficiency	Open-weight
Benchathon	Long but shallow	10	208	8.0%	0.0%	3.9%
	Short / missing Subsumtion	2	208	1.3%	0.0%	1.0%
	Wrong-norm framing	13	208	0.0%	3.3%	11.7%
	Correct outcome, weak Subsumtion	0	208	0.0%	0.0%	0.0%
	Wrong outcome, sound Subsumtion	0	208	0.0%	0.0%	0.0%
ZJS	Long but shallow	1198	8031	23.7%	4.6%	12.0%
	Short / missing Subsumtion	73	8031	0.8%	0.2%	1.2%
	Wrong-norm framing	611	8031	3.2%	7.1%	10.7%
	Correct outcome, weak Subsumtion	19	8031	0.1%	0.4%	0.3%
	Wrong outcome, sound Subsumtion	19	8031	0.5%	0.1%	0.1%

Table 26: Decision-centric error modes on the Doctrinal Principles corpus (4-dimension rubric) under the primary judge.

Mode	Count	n	Flagship	Efficiency	Open-weight
Wrong decision	1541	6365	19.1%	26.9%	26.7%
Correct decision, weak Subsumtion	175	6365	1.0%	2.1%	4.2%
Wrong decision, sound Subsumtion	222	6365	4.4%	3.6%	2.8%

Table 27: Same-base-model reasoning toggle: Qwen3-235B-A22B Thinking-2507 (reasoning on; main evaluation) vs. Instruct-2507 (reasoning off; ablation) under identical prompts, decoding, provider, and primary judge. Pass = judge raw $\geq 50/100$. Latency, output length, and per-generation cost from logged per-call metadata.

Corpus	Variant	n	Judge raw	Pass	Trunc.	Latency s	Words	$/gen
Benchathon	Thinking-2507	28	41.3	18%	0	125.1	678	$0.0144
	Instruct-2507	30	37.8	10%	0	123.6	1356	$0.0005
Doctr. Princ.	Thinking-2507	531	67.3	69%	0	46.1	72	$0.0030
	Instruct-2507	531	63.3	67%	0	8.7	78	$0.0001

Term	**Gloss**
Falllösung	Written case solution – the end-to-end work product that is graded.
Gutachtenstil	Appraisal style: the conclusion follows the reasoning; the register graded here.
Urteilsstil	Judgment style: result first, reasoning after; used in judicial decisions.
Obersatz	Step 1 of the scaffold: the norm hypothesis to be examined.
Definition	Step 2: interpretation of the norm's elements.
Subsumtion	Step 3: anchoring the concrete facts under the norm's elements; broadly, the entire norm-application method.
Ergebnis	Step 4: the resulting legal consequence.
Musterlösung	Expert reference solution against which submissions are graded.
Staatsexamen	State law examination; gate to all regulated legal professions in Germany.
ausreichend	"Sufficient" – lowest passing grade (4–6 of 18 points).
vollbefriedigend	"Fully satisfactory" (10–12 points); customary distinction tier (*Prädikat*).
Referendar	Legal trainee in the two-year practical stage between the state examinations.
Volljurist	Fully qualified jurist (passed both state examinations).
Grundprinzipien	Core doctrinal principles; German name of the Doctrinal Principles corpus.
Bereich(e)	Legal area(s): civil, public, criminal law.
Ja/Nein	Yes/no decision label on Doctrinal Principles items.
Ergebnisrichtigkeit	Rubric dimension: result correctness.
Vollständigkeit	Rubric dimension: issue identification / completeness.
Rechtsgrundlagen	Rubric dimension: legal grounding (norm citation).
Rechtskenntnis	Rubric dimension: doctrinal knowledge.
Schwerpunktsetzung	Rubric dimension: problem depth at the case's focal issues.
Methodischer Stil	Rubric dimension: methodological structure (scaffold execution).
Gliederung	Rubric dimension: organization.
Sprache	Rubric dimension: terminology and language.
Formalia	Rubric dimension: formal correctness.

Table 28: German terms used in the paper, with English glosses.